

Chapter 24. Mechanical Ventilation during General Anesthesia

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Mechanical Ventilation during General Anesthesia: Introduction

In the early years following the first demonstration of anesthesia with an inhalation device, different inhalation anesthetic agents were developed and tested in spontaneously breathing subjects. However, the increasing complexity of surgical procedures, the discovery and use of neuromuscular blocking as well as intravenous anesthetic agents, and the need to protect the airways resulted in the need for mechanical ventilation during general anesthesia. Accordingly, anesthesia ventilators were developed to match the particular needs of general anesthesia, which differ from those in other settings, such as the intensive care unit (ICU). Moreover, modern anesthesia is not restricted to the intraoperative period and mechanical ventilation may be required preoperatively and postoperatively.

This chapter provides a comprehensive review of aspects related to the principles and practice of mechanical ventilation during general anesthesia, as well as during the perioperative period.

Respiratory Effects of Anesthesia and Surgery

General anesthesia has several effects on respiratory function, most notably on control of breathing and the activity of respiratory muscles, which influence the distributions of ventilation and perfusion within the lungs. Moreover, surgery-related positioning of patients, as well as manipulation or displacement of intraabdominal and thoracic organs may further affect ventilation and perfusion, with deterioration in pulmonary gas exchange.

Control of Breathing and General Anesthesia

Objectives of the Control of Breathing

The main objective of control of breathing is the maintenance of blood gases, especially of arterial carbon dioxide (PaCO_2), which is kept within a relatively restricted range. Even a small increase in PaCO_2 may result in a significant increase in minute ventilation. The output of the respiratory center is less sensitive to reductions in partial pressure of arterial oxygen (PaO_2), but hypoxia may enhance the response to PaCO_2 .¹ Another objective of the control of breathing is to maintain the brain pH. Because CO_2 is highly diffusible across the blood–brain barrier, this is achieved through modulation of alveolar ventilation, which regulates PaCO_2 . Last but not least, control of breathing aims at optimizing breathing frequencies and maximizing the work output of the respiratory muscles to maintain adequate gas exchange.²

Control of Breathing

The respiratory rhythm originates from complex neuronal network interactions throughout the nervous system.³ This network consists of smaller and larger networks that interact in generating different breathing patterns, including the regular breathing (eupnea), but also sighs and gasping. The group of neurons responsible for the control of breathing is dispersed across a region in the brainstem termed the *respiratory centers*. These centers are located bilaterally in the reticular formation of the medulla oblongata and pons, beneath the floor of the fourth ventricle, and are grouped into inspiratory, pneumotaxic, and expiratory areas.⁴

The inspiratory area is located in the dorsal part of the medulla and generates rhythmic cycles that are transmitted to the diaphragm, upon contraction of which alveolar ventilation takes place. Impulses originating in peripheral and central chemoreceptors (e.g., in carotid bodies and in the brainstem, respectively), and lungs are transmitted through the vagus and glossopharyngeal nerves to the inspiratory area, downregulating its activity in a negative-feedback mechanism.⁵ Impulses from the pneumotaxic center located in the pons inhibit the inspiratory area continuously, contributing to avoidance of overinflation and modulation of respiratory rate.⁶ Normally, exhalation is a passive process resulting from the elastic recoil of the lungs and chest wall, and the expiratory center located in the ventral portion of the medulla is silent. This center, however, may be activated for increased alveolar ventilation, leading to recruitment of thoracic and abdominal muscles and forced expiration. In addition, alveolar ventilation is regulated by the cerebral motor cortex during voluntary control of breathing, and impulses bypass the respiratory centers being transmitted directly through corticospinal to respiratory motoneurons.⁷

Effects of Anesthesia and Analgesia on the Control of Breathing

Most drugs used to achieve hypnosis, sedation, and relief or suppression of the response to pain affect the control of breathing. Such drugs, which usually result in depression of the alveolar ventilation, have their effects mediated by direct interference with peripheral and central chemoreceptors, changes in behavioral control, suppression of the function of the motoneurons and respiratory muscles, and by general depression of the respiratory center.⁸

Volatile halogenated anesthetic drugs like halothane, isoflurane, enflurane, desflurane, and sevoflurane reduce the response of the respiratory center to hypoxia even at subanesthetic concentrations.^{9–13} In addition, volatile anesthetics can potentiate the effects of neuromuscular blocking agents on the respiratory muscles in a dose-dependent and time-dependent fashion.¹⁴ Intravenous anesthetic drugs like midazolam and propofol also blunt the response to hypoxia at subhypnotic concentrations,^{15,16} while hypnotic concentrations of barbiturates are necessary for decreasing the ventilatory response to hypoxia.¹⁷ Although anesthetics usually depress the respiratory center activity, subhypnotic concentrations of etomidate stimulate ventilation,¹⁸ while during infusion of ketamine, the response to hypercapnia is fairly well maintained¹⁹ or even increased.²⁰

Opioids represent a mainline therapy for analgesia and anesthesia. These drugs, however, can reduce and make the respiratory rate irregular,²¹ resulting both in hypercapnia and hypoxemia. Such effects are mediated mainly by opioid receptors expressed in the peripheral chemoreceptors.²² When infused rapidly as boluses, opioids have the potential to abolish the respiratory center activity completely without producing unconsciousness. Furthermore, opioids can increase the resistance of the upper airway by enhancing the discharge activity in the recurrent laryngeal nerve, and reduce the compliance of the respiratory system through increases in tonic discharges from the expiratory area of the respiratory center.²³

Even when administered in the intrathecal or epidural space, or intramuscularly, opioids have the potential to induce respiratory depression in approximately 1% of patients.²⁴ Respiratory depression induced by epidural opioids results either from systemic absorption, when delayed, or from cephalad migration through the cerebrospinal fluid, when it immediately follows the administration of the drug. Factors that magnify respiratory depression following intrathecal and epidural opioids are: (a) use of intravenous anesthetic drugs; (b) coughing, which may contribute to spread of the drug within the cerebrospinal fluid; (c) accumulation through higher and/or repeated doses; (d) use of morphine; (e) advanced age; (f) lack of opioid tolerance; (g) thoracic epidural placement; (h) general anesthesia; and (i) increased thoracic pressure.²⁴

Pattern of Contraction of Respiratory Muscles and General Anesthesia

Inspiration

Depression of inspiratory muscle activity during anesthesia occurs not in a global but rather in a selective manner. The intravenous administration of the barbiturate thiopentone decreases the mean activity of the genioglossus, sternothyroid and sternohyoid (strap muscles), and scalene muscles in patients.²⁵ Furthermore, tonic pattern of activity of the strap and scalene muscles disappears and those muscles become synchronized with inspiration.

Although diaphragmatic activity is fairly well preserved, the intercostal muscles become silent at concentrations between 0.2 and 1.0 of the minimal alveolar concentration.^{26,27} As a result, a pattern of breathing originates where diaphragmatic activity expands the lower ribcage and the resulting negative intrathoracic pressure moves the upper ribcage inwards. Such pattern can be even more pronounced if the resistance of the upper airways is increased, for example, when narrow endotracheal tubes are used. In fact, it is conceivable that increased resistance solely determines the limited expansion of the ribcage during inspiration. It has been namely shown that one minimal alveolar concentration isoflurane does not reduce ribcage motion, unless hyperpnea occurs.²⁸ Accordingly, ribcage motion during inspiration is not depressed by anesthesia with ketamine.²⁹

Expiration

In contrast to the inspiratory muscles, the expiratory muscles are usually inactive in awake subjects, becoming active during anesthesia. In patients anesthetized with halothane, Warner et al²⁶ attributed active expiration to the motion of the internal intercostal muscles. In fact, most anesthetic agents have been implicated with active expiratory muscle activity.

The purpose of active expiration is not well determined, and its contribution in reducing functional residual capacity (FRC) during anesthesia is controversial.^{30,31}

Functional Residual Capacity during General Anesthesia

Reductions in FRC of as much as 20% during general anesthesia have been reported in healthy individuals and can be larger in the presence of pulmonary comorbidities and obesity.³² Such deterioration occurs early after induction of anesthesia and is stable over time.³³ Also, spontaneous breathing or increases in airway pressure cannot easily reverse a reduced FRC, which persists for hours after recovery from anesthesia.³² Various mechanisms explain the reduction of FRC during anesthesia: (a) changes in the chest wall; (b) changes in the diaphragmatic shape and position; and (c) increase in thoracic blood volume.

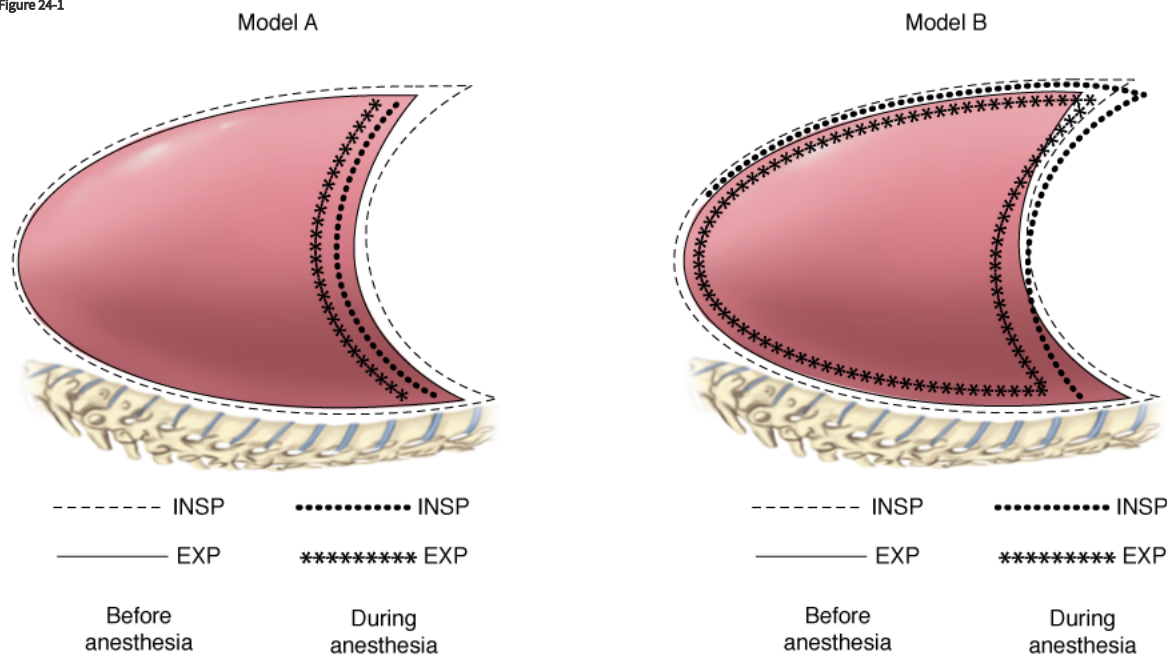
Changes in the Chest Wall

The loss of inspiratory muscle tone associated with anesthesia results in reduced outward recoil of the chest wall, and of the internal diameters of the ribcage,³² leading to changes of approximately 200 to 300 mL in lung volume.^{34–37} In addition, an increase in the anterior curvature of the border of vertebral bodies consistent with increased spinal curvature has been detected.³⁷

Changes in the Diaphragm Shape and Position

Loss of muscle tone during induction of anesthesia may favor a cephalad movement of the diaphragm secondary to pressure of the intraabdominal organs. Although such a mechanism hypothesis has been supported by some authors,^{34,38} it has been challenged by others. Using a dynamic spatial reconstructor technique, Warner et al³¹ showed that it is not the position, but the shape of the diaphragm, that is altered during anesthesia. They³¹ postulated that the most dorsal part of the diaphragm shifts cephalad, while the most ventral part shifts only minimally, or even caudally. Figure 24-1 schematically illustrates the concepts of change in diaphragmatic position and shape.

Figure 24-1



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Models of chest wall and diaphragmatic displacement during general anesthesia. In model **A**, a cephalad shift of the diaphragm is postulated;³⁴ in model **B**, the displacement of the ventral part of the diaphragm is caudal, while the dorsal part shifts cranially.³¹ Both models are associated with a decrease in functional residual capacity (FRC). *Exp*, expiration; *Insp*, inspiration.

Increase of Thoracic Blood Volume

Increases in intrathoracic blood volume resulting from redistribution of blood from the periphery could theoretically further reduce FRC, although this mechanism is controversial.^{34,39}

Atelectasis Formation during General Anesthesia

The concept of atelectasis formation during general anesthesia was first proposed by Bendixen et al⁴⁰ in 1963 to explain a progressive decrease in oxygenation and compliance. Those authors, however, could not confirm that hypothesis using chest radiographs. In addition, other authors were not able to reproduce those findings, observing a prompt rather than gradual change in respiratory mechanics under anesthesia.³³

Collapse of lung zones during anesthesia was first confirmed using computed tomography,⁴¹ and it has been estimated that 90% of patients develop atelectasis under general anesthesia.⁴² Several mechanisms have been suggested to explain the formation of atelectasis during anesthesia: (a) collapse of small airways; (b) compression of lung structures; (c) absorption of intraalveolar gas content; and (d) impairment of lung surfactant function. These mechanisms are not mutually exclusive and may interact with each other.

Collapse of Small Airways

A reduction in FRC below the closing capacity during anesthesia can result in collapse of lung units.⁴³ Two studies documented a decrease of closing capacity and FRC in anesthetized patients,^{44,45} but this finding is not indisputable.⁴⁶ In the presence of increased smooth muscle tone of the small airways, collapse can occur even at FRC values above the closing capacity, for instance in patients with chronic obstructive pulmonary disease (COPD) and asthma, or those receiving anesthetic drugs that promote histamine release and cause bronchoconstriction (e.g., thiopental, atracurium, d-tubocurarine).

Compression of Lung Structures

In the supine position, there is a ventral–dorsal gradient in the transpulmonary pressure.^{47,48} This gradient is determined by the shape of the chest wall and lungs, as well as by lung height multiplied by the tissue density, which may promote atelectasis. Similarly, because the lung area under the heart is greater in the supine

than in the prone position,⁴⁹ the weight of the heart could theoretically favor atelectasis during general anesthesia. Furthermore, the pressure of intraabdominal organs can be transmitted in part to the lung structures in the absence or reduction of diaphragmatic muscle tone.⁵⁰ In addition, increases in intrathoracic blood volume resulting from redistribution of blood from the periphery could theoretically further reduce FRC but this mechanism is controversial.^{34,39}

Absorption of Intraalveolar Gas Content

In the presence of an increased ventilation-to-perfusion ratio, high alveolar oxygen concentrations can lead to a progressive loss of alveolar total gas volume. A computed tomography study showed that 16% to 20% of lung tissue develops collapse or is poorly aerated in patients under general anesthesia.⁵¹ Absorption atelectasis is most relevant when small airway closure occurs, as for instance following reduction of the FRC beyond the closing capacity. The pulmonary circulation then takes up the trapped gas enriched with oxygen, resulting in atelectasis. This mechanism is supported by the fact that lower oxygen fractions at induction of anesthesia limit the development of intrapulmonary shunt.⁵² Mathematical modeling of gas exchange suggests that absorption atelectasis achieves a maximum as fast as 9 minutes if preoxygenation with 100% oxygen is performed over 3 minutes.⁵³

Impairment of Lung Surfactant Function

An early study implicated the volatile anesthetic agents halothane and chloroform in the reduction of lung volume of excised lung dogs. In rabbit lungs, the combination of halothane with higher inspiratory oxygen concentrations increased the permeability of the alveolar–arterial barrier, possibly because of additional effects on the lung surfactant. Also in patients without pulmonary disease, the inhalation of volatile anesthetics reduced the surface tension of tracheobronchial secretions.⁵⁴ Furthermore, inactivation of lung surfactant may occur as a result of large changes in alveolar surface area, which are associated with surfactant aggregate conversion.⁵⁵

General Anesthesia and Distribution of Ventilation and Perfusion

Chapter 37 discusses the effects of mechanical ventilation on the distribution of alveolar ventilation ($\dot{V}_A$) and perfusion ($\dot{Q}$) in detail. Basically, mechanical ventilation is associated with $\dot{V}_A/\dot{Q}$ mismatching and increased intrapulmonary shunt, while diffusion limitation to O₂ plays a marginal role. General anesthesia and surgery have the potential to aggravate the deterioration of gas exchange by further impairment of $\dot{V}_A$ and diversion of perfusion from better ventilated lung zones.

Induction of anesthesia with certain barbiturates (thiopental and thiamylal⁵⁶), opioids (morphine⁵⁷), and neuromuscular blocking agents (atracurium and D-tubocurarine⁵⁸) induces histamine release by mast cells, possibly resulting in bronchoconstriction. Also, endotracheal intubation can elicit an increase of the smooth muscle tone of smaller peripheral airways.⁵⁹ In contrast, volatile anesthetics, including halothane, isoflurane, sevoflurane, and desflurane, exert bronchodilatory effects.⁶⁰

Patient positioning and the surgical procedure itself may affect the distributions of $\dot{V}_A$ and $\dot{Q}$. During anesthesia in the supine position, a cephalad displacement of the diaphragm with reduction in FRC and impairment of $\dot{V}_A$ in juxtadiaphragmatic zones is commonly observed.⁶¹ Some authors, however, suggest that only the dorsal part of the diaphragm is shifted cranially, while the ventral part is displaced caudally.³¹ Such a displacement pattern has been attributed to anesthesia-induced changes in other chest wall structures, like the ribcage, where the diaphragm attaches. On the other hand, surgical procedures requiring the use of the Trendelenburg position and capnoperitoneum, which have more pronounced effects on the respiratory function than the supine position and open abdominal surgery,⁶² may lead to a displacement of the whole diaphragm. Interestingly, however, increased pressure in the lung base during the combination of the Trendelenburg position and laparoscopy also shifts $\dot{Q}$ to lung apical zones, resulting in a preserved $\dot{V}_A/\dot{Q}$ matching, despite deterioration in respiratory mechanics.⁶³ During upper abdominal surgery, the use of retractors also importantly impair lung compliance⁶⁴ and their effects on $\dot{V}_A$ and $\dot{Q}$ are probably the same as those of laparoscopic surgery.

One-Lung Anesthesia and Distribution of Ventilation and Perfusion

One-lung anesthesia has singular effects on $\dot{V}_A$, $\dot{Q}$, and $\dot{V}_A/\dot{Q}$ matching. Secondary to the collapse of the operated lung, $\dot{V}_A$ is limited to a single lung during such procedures. Gravity, however, slightly favors a shift of $\dot{Q}$ toward the dependent lung,⁶⁵ resulting in a better-preserved gas exchange during one-lung anesthesia in the semilateral and lateral decubitus than in the supine position.⁶⁶ Furthermore, hypoxic pulmonary vasoconstriction contributes to further redistribute the blood flow toward the ventilated lung. Theoretically, anesthesia has the potential to interfere with hypoxic pulmonary vasoconstriction. In dogs, volatile anesthetics cause pulmonary capillary dilation that may blunt the redistribution of blood flow and worsen oxygenation.⁶⁷ The use of volatile anesthetics in patients, however, especially at the minimal alveolar concentration, has not been associated with worsening of oxygenation when compared to intravenous anesthetics.^{68–70} On the other hand, the use of epidural anesthesia in combination with intravenous anesthetics seems to result in better oxygenation by mechanisms not related to hypoxic pulmonary vasoconstriction, for instance, reduction of cardiac output.⁷¹

Assistance of the Respiratory Pump during General Anesthesia

General anesthesia does not necessarily require controlled mechanical ventilation. In fact, for more than a century after its description in 1846, volatile anesthetics have been administered to spontaneously breathing patients.⁷² General anesthesia, however, may be associated with a reduction in FRC, tidal volume, changes in

breathing pattern, and alterations in the mechanical properties of the respiratory system, resulting in impairment of oxygenation secondary to $\dot{V}_A/\dot{Q}$ mismatching. Furthermore, the need to protect the airways from aspiration by means of endotracheal intubation and the use of neuromuscular blocking agents to facilitate surgery and endotracheal intubation itself led to the regular use of controlled mechanical ventilation to maintain an adequate gas exchange. Nevertheless, certain surgical procedures can be performed without neuromuscular blocking agents, and endotracheal intubation is not always necessary. In this context, facemasks, laryngeal masks, and other supraglottic devices have largely replaced endotracheal tubes. Accordingly, the interest of anesthesiologists in maintaining spontaneous breathing activity during general anesthesia has been increasing.

In view of these facts, anesthesiologists must be able to partially and fully assist the respiratory pump. For this purpose, several assisted spontaneous breathing and controlled mechanical ventilation modes that are present on ICU ventilators are required. Anesthesia ventilators, however, must be able also to deliver inhalation anesthetics in an environmental and economically friendly way, which is not feasible with semiopen systems. [Table 24-1](#) lists the ventilator modes commonly used during anesthesia in the operating room. Obviously, not all anesthesia ventilators offer all of those modes and some of them are not available on anesthesia machines, as they are currently not appropriate for use in combination with volatile anesthetics—for example, modes based on high-frequency ventilation. However, developing technologies may overcome this limitation in the near future.

Table 24-1: Ventilator Modes Used for Assisting the Respiratory Pump during General Anesthesia

Mode	Comment	Possible Indications in Anesthesia
CPAP (continuous positive airway pressure)	Increases mean airway pressure and allows spontaneous breathing	Induction of anesthesia to avoid major decrease of FRC; decrease inspiratory effort in the presence of an endotracheal tube; increase tidal volume during decreased inspiratory drive; before extubation to avoid loss of the FRC
PSV (pressure-support ventilation)	Increases the pressure at the airway above the expiratory pressure in response to an inspiratory effort; can be triggered either by flow or pressure	Maintenance of adequate ventilation in presence of reduced inspiratory drive, restrictive lung disease, partial neuromuscular blockade or neuromuscular disease; increased inspiratory effort in presence of a narrow airway device; reduction of atelectasis, improved gas exchange, decreased level of sedation; ⁷³ weaning from controlled ventilation
V- or PACV (volume or pressure assist-control ventilation)	Guarantees that the desired tidal volume or driving pressure is delivered upon an inspiratory effort	Allow patients with respiratory drive to control the respiratory rate and/or tidal volume
SIMV (synchronized intermittent mandatory ventilation; also SIMV-VC and SIMV-PC)	With SIMV, if an inspiratory effort is sensed either by pressure or flow during a sensitive time window, the ventilator applies a preset tidal volume (VC) or pressure (PC); a minimum mandatory ventilation can be set, which will guarantee acceptable gas exchange. SIMV has greater acceptance in the operating room than in the ICU	Maintenance of adequate ventilation in presence of inspiratory effort; emergence from anesthesia
VCV (volume-controlled ventilation)	VCV is time-cycled and volume-targeted; PCV is time-cycled and pressure-limited; corresponds to the default mode on most anesthesia ventilators	Tight control of tidal volume in the absence of inspiratory effort, allows better control of partial pressure of end-tidal carbon dioxide ($P_{ET}CO_2$)
PCV (pressure-controlled ventilation)	PCV is time cycled and pressure-limited	Tight control of inspiratory peak pressure; possibly of value in combination with supraglottic airway devices to avoid gastric insufflation ⁷⁴ and upper airway surgery to avoid anastomose insufficiency. Also useful with uncuffed endotracheal tubes to compensate for leaks
VCV+AutoFlow	Used in the volume-controlled ventilation mode, autoflow automatically adjusts the inspiratory flow to achieve the desired tidal volume with lowest possible inspiratory pressure, resulting in an inspiratory decelerating flow pattern. It is also known as pressure-regulated volume-controlled ventilation (PRVCV) and is similar to pressure control with volume guarantee (PC-VG)	In combination with a supraglottic airway or surgery of the upper airways to reduce peak pressures and avoid gastric insufflation and airway anastomose insufficiency
APRV (airway pressure release ventilation)	Time-cycled ventilation with a higher continuous airway pressure that is maintained for longer periods of time (usually >2 s) and periodically released until positive end-expiratory pressure (PEEP) for short periods of time (usually <1 s, typically 0.5 s), resulting in an inverse inspiratory-to-expiratory timing (I:E) ratio. Free (nonassisted) spontaneous breathing activity is possible at any time at the higher airway pressure	Mainly for patients with acute lung injury and/or acute respiratory distress syndrome to increase mean airway pressure and functional residual capacity; may be helpful to limit the pressure in the airways at comparatively higher levels, while allowing spontaneous breathing (e.g., in combination with supraglottic airway devices) and promoting CO ₂ washout
BIPAP (biphasic [intermittent] airway pressure ventilation)	Works similarly to APRV, changing the airway pressure between two levels but allowing more time for expiration than inspiration. BIPAP can be used with time windows that sense inspiratory and expiratory efforts and in combination with pressure-supported breaths at the lower airway pressure level. Also known as BiPAP (bi-level positive airway pressure). A variant called Bi-Vent also provides pressure support for spontaneous breaths at the higher airway pressure level	Similar indications as for PCV and PSV; smoother transition from controlled to assisted spontaneous breathing; maintenance of adequate ventilation in presence of reduced inspiratory drive, restrictive lung disease, partial neuromuscular blockade, or neuromuscular disease

Mode	Comment	Possible Indications in Anesthesia
HFJV (high-frequency jet ventilation)	Usually with respiratory frequencies of 100 to 150 breaths/min and pressure of 1.5 to 2.5 bars; useful with infraglottic, supraglottic, transtracheal, and transluminal airway devices	Surgery and diagnostic procedures of the upper respiratory tract, ventilation of the dependent lung during one-lung anesthesia, emergency transtracheal ventilation ⁷⁵
HFOV (high-frequency oscillatory ventilation)	Usually with respiratory frequencies of 600 to 900 breaths/min and increased mean airway pressure; active instead of passive expiration; requires higher bias gas-flow rates. The resulting tidal volume is relatively low and, in case of CO ₂ retention, can be increased by reducing the respiratory frequency	Ventilation of the dependent lung during one-lung anesthesia, CT-guided percutaneous procedures of the kidneys to reduce movement artifacts, thoracic and abdominal neonatal surgery where an increase of mean airway pressure is indicated (e.g., congenital diaphragmatic hernia, esophageal atresia, correction of abdominal wall defects) ⁷⁶

Ventilator-Associated Lung Injury during Anesthesia

The main goals of mechanical ventilation during general anesthesia are to oxygenate arterial blood and secure adequate CO₂ elimination.⁷⁷ To achieve those aims, tidal volumes as high as 12 to 15 mL/kg of predicted body weight for two-lung ventilation, and 8 to 10 mL/kg for one-lung ventilation have been advocated and represent common practice. It was recognized early that the use of high tidal volumes helped in avoiding intraoperative lung atelectasis⁴⁰ and decreased FRC. Accordingly, the use of positive end-expiratory pressure (PEEP) has been primarily linked to its potential for increasing FRC and avoiding airway closure, even though PEEP does not necessarily improve oxygenation during general anesthesia.^{78,79} Because of the potential harmful effects of PEEP, however, especially on hemodynamics, its use during anesthesia does not represent the standard of care.

There is unequivocal evidence that mechanical ventilation can worsen injury in previously damaged lungs. On one hand, high tidal volumes may promote overdistension of the alveoli (volutrauma). On the other hand, the lack of PEEP can lead to destabilization of lung units at end-expiration, resulting in repeated collapse and opening during tidal ventilation (atelectrauma). The mechanical stress caused by volutrauma and atelectrauma can be translated into the activation of the inflammatory cascade, culminating in the release of proinflammatory mediators by cells, a phenomenon known as mechanotransduction and that represents the basis of the so-called biotrauma. The exact mechanisms by which mechanical stress is sensed and then transduced into inflammation are not fully elucidated, but activation of stretch-sensitive channels, partial cell membrane disruption, and conformational changes in membrane-associated molecules and cytoskeletal structure have been implicated in the gene expression and production of cytokines.⁸⁰ Following, and most probably also simultaneously with the proinflammatory response, the remodeling process can be activated, with upregulation of gene expression and synthesis of collagen types I and III fibers by fibroblasts.^{81–83} Furthermore, the mediators involved in repair and remodeling can also trigger the expression of metalloproteinases,^{84,85} enzymes that promote the degradation of components of macromolecules of the extracellular matrix, namely proteoglycans and glycosaminoglycans, as well as fibrous (collagen and elastin) and structural or adhesive proteins (fibronectin and laminin).⁸⁶ In addition, the mechanical stress generated by high tidal volumes also elicits changes in the expression of some proteoglycan components of the extracellular matrix, possibly changing its mechanical properties.⁸⁷ In fact, fragmentation of pulmonary glycosaminoglycans has been detected with tidal volumes as low as 8 mL/kg in healthy rats, resulting in decreased lung compliance and edema.⁸⁸

Although mechanical stress per se is able to activate the proinflammatory and remodeling responses in noninjured lungs, tidal volumes as high as 22 mL/kg do not exceed the threshold of stress and/or strain necessary to cause lung weight gain, systemic inflammation, and multiple organ dysfunction in healthy pigs.⁸⁹ In line with these findings, nonprotective mechanical ventilation with high tidal volume and zero PEEP did not increase the release of inflammation mediators in surgical patients without lung disease.^{90–92} Such reports support the hypothesis that ventilator-associated lung injury occurs only if the lungs are predisposed to injury by a first hit, that is, an inflammatory process already present in the lungs (the two-hit hypothesis). In this context, inappropriate mechanical ventilation would represent the second hit that elicits lung injury. The two-hit hypothesis, however, has been challenged. In patients undergoing one-lung anesthesia, protective mechanical ventilation with low tidal volume and PEEP decreased the proinflammatory systemic response after esophagectomy.⁹³ Also during two-lung anesthesia, the use of low tidal volume and PEEP attenuated the levels of interleukin (IL)-8 and myeloperoxidase in bronchoalveolar lavage fluid in patients without preexisting lung injury.⁹⁴ Furthermore, in patients admitted to the ICU, large tidal volumes represent a major risk factor for the development of acute lung injury (ALI).^{95,96} Possibly, cytokines released by surgical trauma and surgery-related conditions, for instance, ischemia or reperfusion of different organs, as well as subclinical lung inflammation, could function as a first hit, and thus explain those discrepancies.

Whether protective mechanical ventilation strategies during anesthesia in patients without lung injury may avoid the onset of ALI, the acute respiratory distress syndrome, or other forms of pulmonary complications remains to be determined.

Postoperative Pulmonary Complications

Approximately 234 million major surgical procedures are performed every year worldwide, with 2.6 million representing high-risk procedures.⁹⁷ About half of the patients undergoing high-risk interventions develop complications and 315,000 die during their hospital stay. Postoperative pulmonary complications are as common as cardiac complications following noncardiac surgery⁹⁸ and occur in approximately 5% of all patients undergoing surgery and anesthesia.⁹⁹ This number,

however, increases importantly in the presence of certain risk factors, including older age, low preoperative oxygen saturation, respiratory infection in the last 30 days preceding surgery, anemia (hemoglobin level <10 g/dL), surgery of the upper abdominal or thorax, surgery lasting longer than 2 hours, and emergency procedures.⁹⁹

The incidence of postoperative pulmonary complications also vary according to their definition,^{98,100–102} and the following entities and/or findings have been classified as a postoperative pulmonary complication: respiratory failure from pulmonary or cardiac origin; pneumonia and respiratory infection; pleural effusion and atelectasis; pneumothorax; bronchospasm; and need for noninvasive respiratory support or reintubation. Once a postoperative pulmonary complication occurs, the average hospital stay is prolonged and the risk of in-hospital death increases.^{99,103}

Different strategies have been proposed to reduce the incidence of postoperative pulmonary complications, including postoperative lung-expansion maneuvers, preoperative intensive inspiratory muscle training, selective rather than routine use of nasogastric tubes, use of short-acting rather than long-acting neuromuscular blockade, and laparoscopic instead of open bariatric surgery.^{98,104,105} Despite the compelling rationale for the use of PEEP (combined or not with lung-recruitment maneuvers) to prevent formation of atelectasis in the intraoperative period, an increase of FRC did not improve outcome following surgery in a nonselected population.¹⁰⁶ The outcome might be different if patients with higher risk for postoperative pulmonary complications were selected. For this purpose, different propensity scores are available, as shown in Table 24-2. The major weakness common to all of those prediction models is that events and interventions in the intraoperative period, for example, bleeding and hemodynamic instability, are not taken into account and may impair their accuracy. Accordingly, the impact of the intraoperative ventilation strategy is not taken into account. In view of these facts, such scores require validation in independent trials.

Table 24-2: Studies on Prediction of Postoperative Pulmonary Complications

Ref	Type of PPC	Predictors of PPCs	Type and Number of Patients/Setting
Christenson 1996 ¹⁰⁷	ARDS (1%)	Hypertension; Diabetes mellitus Smoking Reoperative surgery NYHA functional class 3 or 4 Low left-ventricular ejection fraction (<40%)	Cardiac surgery with cardiopulmonary bypass (<i>n</i> = 3848)
Kutlu 2000 ¹⁰⁸	ALI (0.8%) ARDS (3.1%) Mortality (3.5%)	Age Sex Diagnosis (cancer vs. not) Extent of lung resection	Pulmonary resection (<i>n</i> = 1139)
Tandon 2001 ¹⁰⁹	ARDS (14.5%), pleural effusion (15%), prolonged pneumothoraces (4.1%), hydropneumothoraces (1.8%), empyema (2.4%), chylothorax (2.4%), pneumonia (17.8%), pulmonary embolism (1.8%), laryngeal edema (0.5%)	BMI Smoking Perioperative fluid/transfusion administration Use of inotropes Perioperative hypoxemia Perioperative hypotension Duration of surgery One-lung ventilation Postoperative leak	Elective esophagectomy (<i>n</i> = 168)
Milot 2001 ¹¹⁰	ARDS (0.4%) Total mortality (5.6%) Mortality in ARDS (15%)	Prior cardiac surgery Shock Perioperative fluid/transfusion administration	Cardiac surgery with CPB (<i>n</i> = 3278)
Licker 2003 ¹¹¹	Total PPC (18.5%): ALI (4.2%), ARDS (1.5%), atelectasis (7.1%), prolonged chest drainage (5.0%), pneumonia (4.2%), bronchopulmonary fistula (1.8%), pleural effusion (2%), thromboembolism (0.5%) Mortality (3.0%)	Age History of chronic alcohol consumption Diabetes mellitus FEV ₁ High-risk ASA status Pneumonectomy Extended resections Duration of surgery One-lung ventilation Barotrauma Amount of fluid infused PaO ₂ /FiO ₂ at arrival in ICU	Thoracic surgery for non- small-cell lung carcinoma (<i>n</i> = 879)

Ref	Type of PPC	Predictors of PPCs	Type and Number of Patients/Setting
Fernandez-Perez 2006 ¹¹²	Total PPC (18%): acute lung injury (50%), cardiogenic pulmonary edema (17%), pneumonia (23%), bronchopleural fistula (7%), pulmonary thromboembolism (3%)	Larger intraoperative V _T Perioperative fluid administration	Pneumonectomy (<i>n</i> = 170)
Alam 2007 ¹¹³	Total PPC (5.3%): ALI or ARDS (3.1%), pneumonitis (2.2%)	Perioperative fluid administration FEV ₁ % predicted postoperative	Thoracic surgery, lung resection (<i>n</i> = 1428)
Johnson 2007 ¹¹⁴	Total PPC (3%): pneumonia (35%), systemic sepsis (23%), and cardiac arrest (13.3%)	Higher ASA classification Emergency operation Preoperative sepsis Elevated creatinine	Major general or vascular procedures (<i>n</i> = 180,359)
Canet 2010 ⁹⁹	Total PPC (5%): respiratory infection (1.6%), respiratory failure (2.6%), bronchospasm (1.8%), atelectasis (1.4%), pleural effusion (1.7%), aspiration pneumonitis (0.4%)	Age Male sex Low preoperative SpO ₂ Acute respiratory infection during the month before surgery Preoperative anemia Positive cough test Upper abdominal or intrathoracic surgery Duration of procedure Emergency surgery	Surgical procedures (general, neuraxial, or regional anesthesia) (<i>n</i> = 2464)
Sen 2010 ¹¹⁵	ARDS (7.5%), pneumothorax (6.2%), prolonged air leak (2.1%), empyema (4.2%) Mortality (2.7%)	History of chronic alcohol consumption ASA score Fresh-frozen plasma Pulmonary resection type	Pulmonary resection (<i>n</i> = 143)
Ferguson 2011 ¹¹⁶	PPC (38%)	Age FEV ₁ % predicted DL _{CO} % predicted Performance status Serum creatinine Smoking Transthoracic resection	Esophagectomy (<i>n</i> = 516)
Kor 2011 ¹¹⁷	ALI (2.6%)	Cardiac surgery Vascular surgery Thoracic surgery Diabetes mellitus COPD Gastroesophageal reflux Alcohol abuse	High-risk surgery (<i>n</i> = 4366)

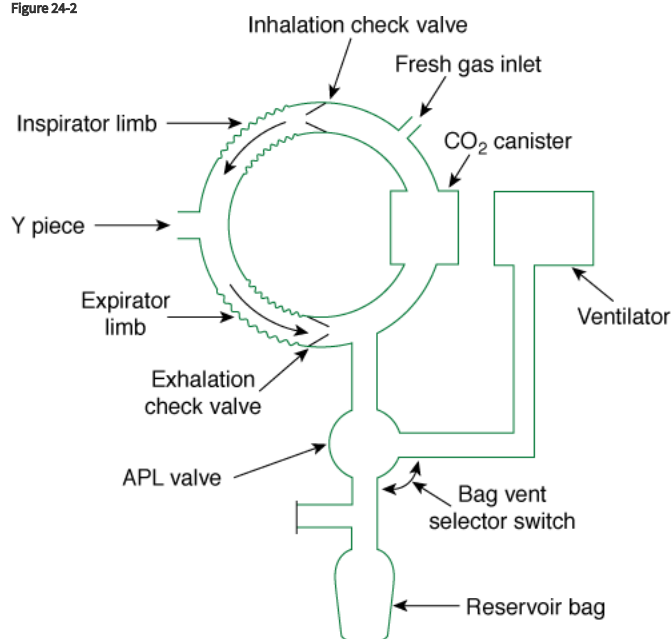
Abbreviations: ALI, acute lung injury; ARDS, acute respiratory distress syndrome; ASA, American Society of Anesthesiologists; BMI, body mass index; COPD, chronic obstructive pulmonary disease; CPB, cardiopulmonary bypass; DL_{CO}, diffusing lung capacity for carbon monoxide; FEV₁, forced expiratory volume at 1 second; ICU, intensive care unit; NYHA, New York Heart Association; PPC, postoperative pulmonary

Anesthesia Ventilators

General anesthesia was first delivered using an open-inhaler system by John Snow as early as 1846. Because anesthesia itself often resulted in depression of the respiratory motor output, ventilation had to be assisted by bag ventilation. Thus, besides vaporizers, anesthetic circuits, and scavenging systems, mechanical ventilators became a fundamental part of the anesthesia machine as a means to automate ventilation. Mechanical ventilators, however, represent nowadays more than simple bag substitutes. Rather, they play a central role in the anesthesia machine and must match different requirements than their counterparts in the ICU, justifying the use of the term *anesthesia ventilator*.

Most commonly, the anesthesia ventilator is used in combination with the so-called circle system, which is depicted in Figure 24-2. That system includes an adjustable airway-pressure limiting valve that can be set by the anesthesiologist to limit airway pressure during bag ventilation, a bag ventilation valve itself with a respective on/off valve, an entrance for a fresh mixture of respiratory and anesthetic gases, a scavenging system, an absorber for CO₂, and the anesthesia ventilator. During the expiratory phase, the anesthesia ventilator is filled with a gas mixture coming from the patient, which is pressed again into the anesthesia circuit during inspiration. In other words, such system permits exhaled air to be rebreathed by the patient, thus reducing the consumption of anesthetic gas agents. In general terms, the anesthesia ventilator is embedded in a semiclosed system in contrast to ICU ventilators, which represent open or semiopen systems.

Figure 24-2



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Schematic of the circle system used in most anesthesia machines. Fresh gas consisting of a mixture of oxygen, nitrogen, and anesthetic gases flows into the circle system between the CO₂ canister, which contains a CO₂ absorber salt, and the inhalation check valve. Gas exhaled by the patient passes an exhalation check valve and refills the anesthesia ventilator and/or reservoir bag. Because the flow rate of fresh gas can be as low as the gas consumption rate, the circle system can recirculate most of the exhaled gas. The anesthesiologist can ventilate the patient manually by bag or using an anesthesia ventilator upon switching of a selector valve. During bag ventilation, the adjustable pressure-limiting (APL) valve is set to limit the maximal pressure in the circuit, while, during function of the anesthesia ventilator, the maximal pressure is usually fixed at 70 cm H₂O by an internal pressure-relief valve. (Adapted from Andrews JJ. Inhaled anesthetic delivery systems. In: Miller RD, ed. *Anesthesia*. New York: Churchill Livingstone; 1994.)

Chapter 2 presents a comprehensive scheme for classification of mechanical ventilators, which also applies for devices embedded in anesthesia machines. The classification of anesthesia ventilators, however, according to: power source, driving mechanism, cycling mechanism, and design, that is, bellows, piston, or turbine, is useful for the anesthesiologist, because it allows inference of expected performance and safety aspects of the anesthesia machine.

Power Source

Some anesthesia ventilators are powered by compressed gas, but most modern ventilators are operated either by electricity alone or a combination of electricity and compressed gas. When compressed gas is used, the ventilator can function even in the absence of electricity.

Driving Mechanism

Several anesthesia machines have high-pressure and low-pressure circuits, that is, are a double-circuit. In the low-pressure circuit, the mixture consisting of anesthetic and respiratory gases are contained in a bag or bellows. Gas at higher pressure, usually oxygen (sometimes mixed with air), compresses the bellows to

move the anesthetic gas mixture into the lungs.

Cycling Mechanism

The cycling mechanism describes how the ventilator initiates inspiration in the control mode. Most contemporary anesthesia ventilators are time cycled and use solid-state electronics for timing. Older devices may be based on pneumatic and fluidic timers.

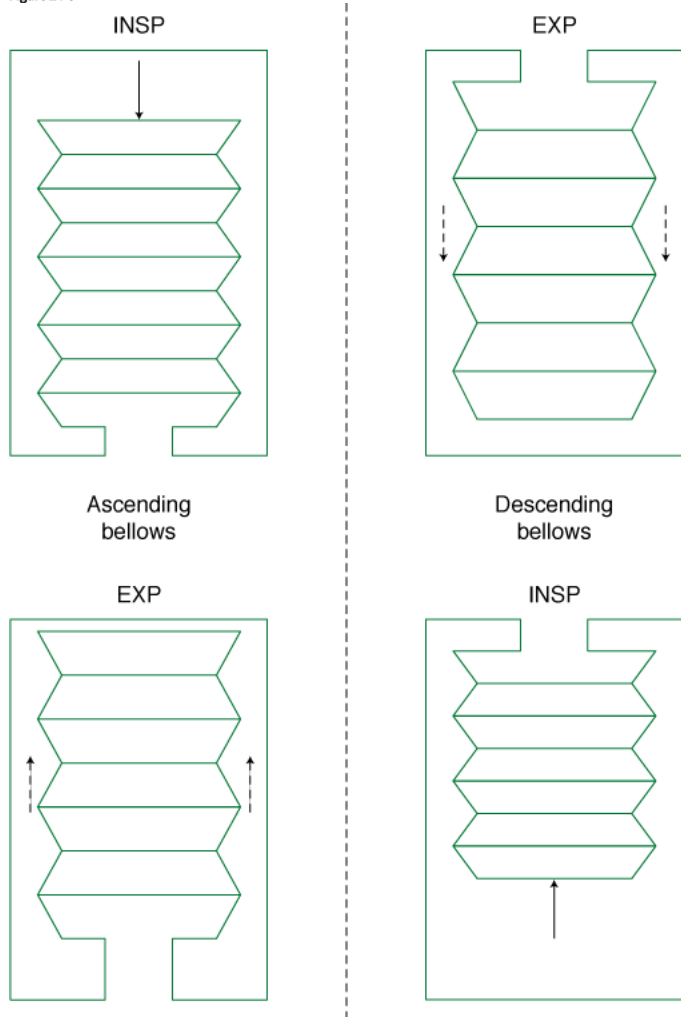
Design

Bellows Design

The first anesthesia ventilators were based on the bellows design. Bellows ventilators more or less simply automate bag ventilation, and can be still found in modern anesthesia machines. Bellows are located in a chamber where a driving gas flows in to generate an external pressure that squeezes out their contents. The driving gas mixture, which usually consists of [oxygen](#) (sometimes mixed with air), is completely separated from the anesthetic gas mixture delivered to the patient.

Bellows are classified according to their motion during expiration, as shown in [Figure 24-3](#). Accordingly, ascending (standing) bellows move upwards, while descending (hanging) bellows move downwards in the expiratory phase. Alternatively, bellows may be displaced horizontally, minimizing the effect of gravity on bellows movement. In addition, horizontal bellows that have a large surface area offer reduced mechanical resistance to movement.

Figure 24-3



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Schematic of anesthesia machine bellows. *Left*: Ascending type; *right*: descending type. *Straight lined arrows* represent the direction of movement during inspiration; *dashed line arrows* represent the direction of movement during expiration. *Exp*, expiration; *Insp*, inspiration.

Given that the weight of a descending bellows promotes its expansion, it may continuously inflate and deflate even in the absence of adequate patient ventilation, for instance during patient disconnection from the anesthesia circuit. Because of this serious safety concern, most bellows ventilators in anesthesia machines are of the ascending type. During inspiration, a driving gas squeezes the bellows downward, pushing its content into the anesthesia circuit. Simultaneously, the driving gas closes a relief valve, avoiding escape of the anesthetic gas mixture into the scavenging gas system. During expiration, the pressure around the bellows is reduced to zero, allowing escape of the driving gas and upward movement of the bellows, which is mainly driven by the pressure in the patient's airways. Because the ventilator relief valve opens first at 2 to 3 cm H₂O during expiration, the bellows are filled first before any gas is scavenged, and a minimal PEEP of 2 to 3 cm H₂O results.

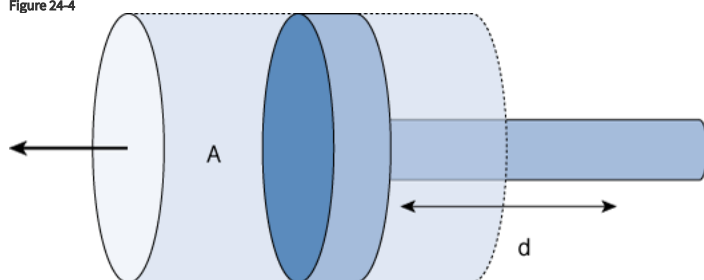
Another problem associated with the bellows anesthesia ventilator is that fresh gas flow is permanently added to the anesthesia circuit during the whole-breath cycle. Accordingly, the effectively delivered tidal volume may differ from the anesthesia ventilator settings depending on inspiration time. Furthermore, the compliance of the breathing circuit may also result in differences between desired and delivered tidal volume.

The problem of fresh gas flow-dependence on tidal volume is not exclusive to bellows anesthesia ventilators and can be fixed by decoupling systems that are present in some anesthesia machines. For instance, the fresh gas can be decoupled from the anesthesia circuit by a valve that closes during inspiration, diverting the fresh gas to the bag reservoir during inspiration. Furthermore, flow and volume sensors placed near the patient–airway interface may be used to compensate for differences between desired and effectively delivered tidal volume (V_T). Such differences may result not only from inappropriate fresh gas flow rates, but also from expansion of the breathing circuit during inspiration. Those sensors, however, are prone to accumulation of moisture, which may impair their accuracy, jeopardizing the performance of the ventilator.

Piston Design

The use of mechanical ventilators based on piston (Fig. 24-4) instead of bellows design in anesthesia machines is increasing. Anesthesia piston ventilators are able to deal more efficiently with volume compensation secondary to compliance of the breathing circuit because they depend not on measurement of airflow or volume but rather pressure, which is less sensitive to the accumulation of moisture or secretions. Furthermore, control of the piston's displacement allows more accurate volume delivery, especially in situations where low tidal volumes are required, for example, in neonatal anesthesia. In addition, the piston ventilator allows a more rapid increase in flow than a bellows ventilator, which is particularly important for pressure-controlled ventilation (PCV).

Figure 24-4



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Schematic of an anesthesia machine piston. A , area; d , displacement.

Because of the ease and accuracy with which the piston can be controlled, advanced ventilation modes can be implemented through software enhancements to the piston ventilator. The basic piston design has proven itself to be a versatile platform for anesthesia ventilator design, which permits transferring differentiated mechanical ventilation modes from the ICU to the operating room.

Turbine Design

An innovative anesthesia ventilator design is the electrically driven and controlled compressor turbine. The turbine has two particularities that make it attractive for anesthesia ventilators:¹¹⁸ (a) it builds up the breathing pressure and delivers the corresponding flow to the patient during inspiratory time; and (b) it delivers a circuit flow that is required to mix the gas within the breathing system independently of patient inspiratory effort. During the inspiratory phase of controlled ventilation, the turbine transports the breathing gas from the breathing bag reservoir to the patient. During expiration, the gas returns to the bag and is additionally circulated and mixed.¹¹⁸ This anesthesia ventilator design is particularly important for closed-systems, which allow minimal consumption of anesthetic gases, that is, quantitative anesthesia. Because a turbine-based ventilator performs as well as the best compressed-gas ICU ventilators,¹¹⁹ it offers a useful platform for implementation of advanced mechanical ventilation modes in anesthesia.

Table 24-3 depicts the basic characteristics of some available anesthesia ventilators. For a complete and comparative description of those devices we recommend the reader to assess the homepage of the nonprofit research agency ECRI Institute at www.ecri.org.

Table 24-3: Basic Characteristics of Some of Commercially Available Anesthesia Ventilators

Fabricant	Dräger Medical	Dräger Medical	Dräger Medical	Dräger Medical
<i>Model</i>	<i>Zeus IE</i>	<i>Primus IE</i>	<i>Fabius GS Premium, Fabius MRI</i>	<i>Cicero EM, Cicero B</i>
Ventilator design	Turbine	Piston	Piston	Piston
Bellows size	—	—	—	—
Ventilation modes*	Manual/spontaneous, CPAP, VCV, Autoflow, PCV , SIMV-PS, PSV	Manual/spontaneous, CPAP, VCV, Autoflow, PCV , SIMV-PS, PSV	Manual/spontaneous, CPAP, VCV, PCV , SIMV, PSV	Manual/spontaneous, CPAP (APL valve), VCV, PCV , SIMV
Tidal volume (mL)	20 to 1500	20 to 1400	20 to 1400	20 to 1400
Minute volume (L/min)	Up to 40	Up to 50	Up to 50	—
Respiratory rate (breaths/min)	Up to 80	3–100	4–60	6 to 80 (VCV, PCV) 3 to 80 (SIMV)
Inspiratory flow (L/min)	180 maximum	150 maximum	10 to 75 in PCV /10 to 85 in PS and SIMV	5 to 75
I:E ratio	4:1 to 1:4	5:1 to 1:99	4:1 to 1:4	1:3 to 2:1
Inspiratory pause (% of T _I)	20 to 50	0 to 60	0 to 50	0 to 60
Pressure limit (cm H ₂ O)	Up to 70	Up to 70	Up to 70	Up to 70
PEEP (cm H ₂ O)	0 to 35	0 to 20 (Pmax-10 hPa in volume mode; Pmax-6 hPa in pressure mode)	0 to 20	0.2 to 15
Control of inspiratory tidal volume	Fresh gas decoupling and compliance compensation	Fresh gas decoupling and compliance compensation	Fresh gas decoupling and compliance compensation	Fresh gas decoupling and compliance compensation
Fabricant	GE Healthcare	MEDEC	Spacelabs Healthcare	
<i>Model</i>	<i>Aisys Carestation, Avance</i>	<i>Neptune, Saturn Evo Color, Saturn Evo Standard</i>	<i>Blease Focus, Blease Serious</i>	
Ventilator design	Bellows (ascending)	Bellows (horizontal bag-in-bottle)	Bellows (ascending bag-in-bottle)	
Bellows size	1500 mL	1 for neonate to adult	Pediatric/adult	
Ventilation modes*	Manual/spontaneous, CPAP, VCV, PCV , PCV-VG, SIMV-VC, SIMV-PC, PSVPro, end-tidal control (not in the <i>Avance</i>)	Manual/spontaneous, VCV, PCV , SIMV, PSV	Manual/spontaneous, VCV, (Precision) PCV , SIMV-PS	
Tidal volume (mL)	20 to 1500	10 to 1600	20 to 1500	

Fabricant	Dräger Medical	Dräger Medical	Dräger Medical	Dräger Medical
Minute volume (L/min)	0.08 to 120	Not specified	0.3 to 25	
Respiratory rate (breaths/min)	4 to 100	4 to 80	2 to 99	
Inspiratory flow (L/min)	1 to 120	Automatic, decelerating flow pattern	0 to 100, variable	
I:E ratio	2:1 to 1:8	4:1, 3:1, 2:1, 1:1, 1:1.5, 1:2, 1:3, 1:4, 1:5, 1:6	2:1 to 1:5	
Inspiratory pause (% of T _I)	0 to 60	0 to 50	0 to 50	
Pressure limit (cm H ₂ O)	12 to 100	7 to 99	10 to 70, adult 10 to 50, pediatric	
PEEP (cm H ₂ O)	4 to 30	0 to 20	3 to 20	
Control of inspiratory tidal volume	Fresh gas decoupling and compliance compensation	Compliance compensation	Fresh gas decoupling and compliance compensation, tidal volume increased by 10% at every tenth breath (sigh function)	

Abbreviations: CPAP, continuation positive airway pressure; PCV, pressure-controlled ventilation; PCV-VG, pressure controlled with volume guarantee; PSV, pressure-support ventilation; SIMV, synchronized intermittent mandatory ventilation; SIMV-PC, SIMV with pressure control; SIMV-PS, SIMV with pressure support; SIMV-VC, SIMV with volume control; VCV, volume-controlled ventilation.

*Ventilation modes listed may be optional.

The authors and the publisher do not take responsibility for the information contained in this table.

Reference: Healthcare Product Comparison System: Anaesthesia Units at www.ecri.org.

Mechanical Ventilation Strategies during General Anesthesia

Current Clinical Practice and Evidence

Mechanical ventilation strategies can influence the development of ventilator-induced lung injury and affect clinical outcome following surgery. Despite a compelling rationale for the use of lower V_T , PEEP, and recruitment maneuvers, the role of protective mechanical ventilation during general anesthesia has not been established. Nevertheless, several clinical trials have addressed the impact of mechanical ventilation on pulmonary function, lung injury, and clinical outcome (Table 24-4).

Table 24-4: Studies on the Impact of Ventilator Settings during General Anesthesia on Pulmonary Function, Lung Inflammation, and Outcome

		Control Group		Protective Group		Inflammation Markers						Intraoperative Outcomes		Postope
Ref	Type and Number Patients/Setting	PEEP	V _T mL/kg/BW	PEEP	V _T mL/kg/BW	TNF α	IL-1	IL-2	IL-6	IL-8	IL-10	compl	oxy	oxy
Lee 1990 ¹²⁰	Postoperative pts (not neurosurgical) (<i>n</i> = 103)/ICU	—	12	—	6	NA	NA	NA	NA	NA	NA	NA	↓	NA
Wrigge 2000 ⁹⁰	Elective surgery (<i>n</i> = 39)/OR	0	15	0	6	=	=	NA	=	NA	=	NA	NA	NA
				10	6	=	=		=		=			
Wrigge 2004 ⁹¹	Thoracotomy (<i>n</i> = 35)/Laparotomy (<i>n</i> = 30)/OR	0	12/15	10	6	=	=	=	=	=	=	=	=	=
Koner 2004 ¹²¹	Cardiopulmonary bypass (<i>n</i> = 44)/OR	0	10	5	10	=	NA	NA	=	NA	NA	NA	↑	↑
				5	6								↑	↑
Wrigge 2005 ⁹²	Cardiopulmonary bypass (<i>n</i> = 44)/ICU	—	12	—	6	↓	NA	NA	=	=	NA	NA	NA	NA
Reis Miranda 2005 ¹²²	Cardiopulmonary bypass (<i>n</i> = 62)/OR-ICU	5	9	10	4/6	=	NA	NA	=	↓	=	NA	↑	↑
Zupancich 2005 ¹²³	Cardiopulmonary bypass (<i>n</i> = 40)/OR	2/3	12	10	8	NA	NA	NA	↓	↓	NA	NA	=	=
Michelet 2006 ⁹³	Esophagectomy (<i>n</i> = 52)/OR	0	9	5	5	=	↓	NA	↓	↓	NA	NA	↑	↑
Choi 2006 ¹²⁴	Surgery for ≥ 5 h (<i>n</i> = 40)/OR	0	12	10	6	Activation of BALF coagulation (increase in TATc, soluble tissue factor and factor VIIa) in the control group						NA	NA	=
Cai 2007 ¹²⁵	Elective excision of intracranial lesion (<i>n</i> = 16)/OR	0	10	0	6	NA	NA	NA	NA	NA	NA	NA	NA	NA
Wolthuis 2008 ⁹⁴	Abdominal surgery with duration ≥ 5 h (<i>n</i> = 46)/OR	0	12	10	6	=	=	=	=	↓	NA	NA	↑	↑
Determann 2008 ¹²⁶	Abdominal surgery with duration ≥ 5 h (<i>n</i> = 40)/OR	0	12	10	6	No differences on levels of biomarkers of lung epithelial injury (BALF and systemic levels of SP-A, SP-D, CC16 and sRAGE)						NA	=	=

		Control Group		Protective Group		Inflammation Markers						Intraoperative Outcomes		Postoperative
Ref	Type and Number Patients/Setting	PEEP	V _T mL/kgIBW	PEEP	V _T mL/kgIBW	TNF α	IL-1	IL-2	IL-6	IL-8	IL-10	compl	oxy	oxy
Determann 2010 ¹²⁷	Medical-surgical patients admitted to ICU (<i>n</i> = 150)/ICU	Best PEEP	10	Best PEEP	6	=	=	NA	↓	NA	NA	=	=	=
Sundar 2011 ¹²⁸	Cardiopulmonary bypass (<i>n</i> = 149)/OR	Best PEEP	10	Best PEEP	6	NA	NA	NA	NA	NA	NA	NA	NA	NA
Yang 2011 ¹²⁹	One-lung ventilation in elective lobectomy (<i>n</i> = 100)/OR	0	10	5	6	NA	NA	NA	NA	NA	NA	NA	NA	↑

Abbreviations: ↑, increased; ↓, decreased; =, unchanged; *atelect*, atelectasis; *BALF*, bronchoalveolar lavage fluid; *CC16*, Clara cell protein; *compl*, compliance; *IBW*, ideal body weight; *ICU*, intensive care unit; *IL*, interleukin; *NA*, not assessed; *OR*, operation room; *oxy*, arterial oxygenation; *PEEP*, positive end-expiratory pressure; *PPC*, postoperative pulmonary complication; *SP*, surfactant protein; *sRAGE*, soluble receptor for advanced glycation end products; *TATC*, thrombin-antithrombin complex; *TNF*, tumor necrosis factor; *V_T*, tidal volume.

The clinical practice of mechanical ventilation may vary widely across countries and even within a single department. In fact, anesthesiologists ventilate noninjured lungs with both high and low tidal volumes, as well as high and low levels of PEEP, because definitive evidence supporting the one or the other approach is still missing.

Tidal Volumes

As shown in Table 24-4, the effects of tidal volumes have been investigated during different surgical conditions, including elective abdominal surgery, cardiopulmonary bypass, and thoracic and esophageal surgery, as well as in patients admitted to the ICU after surgery. The lower V_T used in those studies was 6 mL/kg (ideal body weight), while the higher V_T ranged from 10 to 15 mL/kg. Furthermore, lower V_T were used in combination with higher PEEP (5 to 10 cm H₂O), while high V_T were used in combination with moderate PEEP (2 to 5 cm H₂O).

Both physiologic and clinical parameters have been investigated in those trials. In elective abdominal surgery, the combination of higher V_T and lower PEEP resulted in no differences in pulmonary biomarkers (inflammatory mediators), in two studies,^{90,91} whereas IL-8 in was higher in one study⁹⁴ and bronchoalveolar coagulation was more pronounced in another trial.¹²⁴ In patients undergoing cardiopulmonary bypass, higher V_T were associated with increased lung inflammation in three trials,^{121–123} although one study could not confirm such detrimental effect.⁹² High V_T resulted in elevation of plasma levels of inflammatory mediators in one study on thoracic-esophageal surgery,⁹³ but that finding has been disputed.⁹¹ In postoperative patients recovering in the ICU, lower V_T was associated with decreased plasma levels of IL-6.¹²⁷

With respect to gas exchange, the combination of lower V_T and higher PEEP resulted in improved oxygenation in different trials,^{94,121,122} but also increased PaCO₂ and lowered arterial pH in two investigations.^{94,124}

The number of hospital-free days could be reduced by protective mechanical ventilation with lower tidal volume and higher PEEP in a postoperative mixed medical surgical patients in one study,¹²⁰ but several trials failed to show such a beneficial effect.^{91,93,121,124,127,129} In cardiac surgery patients, the use of a ventilation strategy with V_T of 6 mL/kg resulted in a higher proportion of patients extubated 6 hours after surgery and a lower reintubation rate as compared with a nonprotective strategy of V_T 10 mL/kg.¹²⁸

Positive End-Expiratory Pressure

Table 24-5 depicts selected prospective randomized trials that adjusted the level of PEEP while maintaining V_T approximately constant during open abdominal,^{130–132} laparoscopic,^{133–135} bariatric,^{136–138} and neurosurgery.¹³⁹ Except for one study,¹³⁷ the lower level of PEEP was zero in those trials, while the highest levels

varied between 5 and 12 cm H₂O. In four studies, a recruitment maneuver was performed before setting high PEEP.^{131,133,136,137} In most studies in Table 24-5, higher PEEP was associated with better intraoperative oxygenation, with no major changes in pulmonary inflammation. In three studies, a reduction in atelectasis was also observed in the postoperative period.^{136,138,139} The use of higher PEEP during surgery improved oxygenation in the postoperative period in two studies,^{131,138} while the other four trials observed no benefit.^{132–134,136}

Table 24-5: Studies Employing Randomized-Design Comparisons of Different Positive End-Expiratory Pressure Strategies during General Anesthesia

Ref	Type and Number Patients/Setting	Control Group		Protective Group		Intraoperative Outcomes		Postoperative Outcomes			
		PEEP	V _T mL/kg IBW	PEEP	V _T mL/kg IBW	compl	oxy	oxy	atelect	PPC	Length of Stay
Berthelsen 1979 ¹³⁰	Abdominal surgery (<i>n</i> = 40)/OR	0	—	10	—	NA	NA	=	NA	NA	NA
Neumann 1999 ¹³⁹	Neurosurgery or eye surgery (<i>n</i> = 13)/OR	0	—	10	—	NA	NA	NA	↓	NA	NA
Tusman 1999 ¹³¹	Abdominal surgery (<i>n</i> = 30)/OR	0	7/9	5	7/9	=	=	=	NA	=	NA
				5 + R	7/9	↑	↑	↑	NA	=	NA
Wetterslev 2001 ¹³²	Abdominal surgery (<i>n</i> = 40)/OR	0	—	Best PEEP	—	NA	↑	=	NA	NA	NA
Pang 2003 ¹³³	Laparoscopic cholecystectomy (<i>n</i> = 24)/OR	0	10	5 + R	10	NA	↑	=	NA	NA	NA
Meininger 2005 ¹³⁴	Laparoscopic surgery (<i>n</i> = 20)/OR	0	—	5	-	NA	↑	=	NA	NA	NA
Whalen 2006 ¹³⁷	Laparoscopic bariatric surgery (<i>n</i> = 20)/OR	4	8	12 + R	8	↑	↑	=	NA	=	=
Talab 2009 ¹³⁸	Laparoscopic bariatric surgery (<i>n</i> = 66)/OR	0	8 to 10	5	8 to 10	NA	=	=	↓	↓	=
				10	8 to 10		↑	↑	↓↓	↓↓	↓
Reinius 2009 ¹³⁶	Bariatric surgery (<i>n</i> = 30)/OR	0 + R	10	10	10	↑	=	=	=	NA	NA
				10 + R	10	↑	↑	=	↓		
Kim 2010 ¹³⁵	Laparoscopic cholecystectomy (<i>n</i> = 30)/OR	0	8	5	8	=	↑	↑	NA	NA	NA

Abbreviations: ↑, increased; ↓, decreased; =, unchanged; *atelect*, atelectasis; *compl*, compliance; *IBW*, ideal body weight; *ICU*, intensive care unit; *NA*, not assessed; *OR*, operation room; *oxy*, arterial oxygenation; *PAW*, airway pressure; *PEEP*, positive end-expiratory pressure; *PPC*, postoperative pulmonary complication; *R*, recruitment maneuver; *V_T*, tidal volume.

There is also uncertainty with regard to the effects of PEEP on outcome of surgical patients. Higher PEEP levels contributed to prevent postoperative pulmonary complications in one study,¹³⁸ but two trials reported no effect.^{131,137}

Recruitment Maneuvers

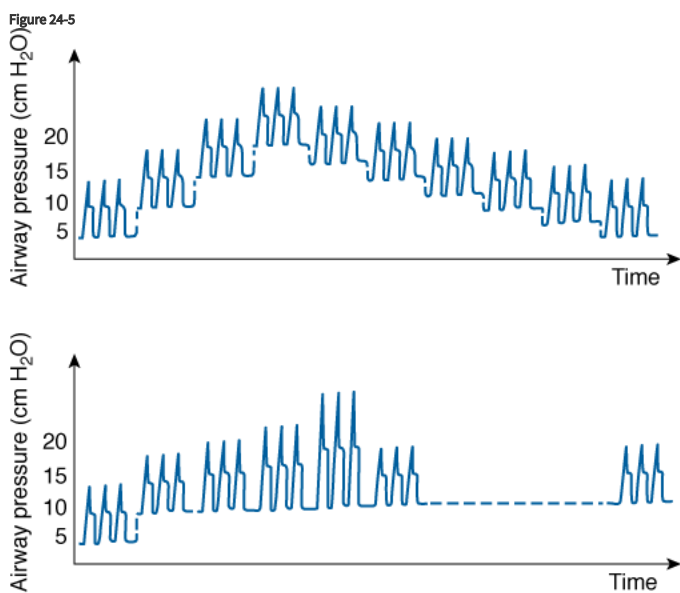
The use of recruitment maneuvers in patients with ALI and/or acute respiratory distress syndrome is controversial. Moreover, alveolar flooding and consolidation of lung parenchyma are common findings in injured lungs, which challenge the rationale for the use of recruitment maneuver. In contrast, the presence of atelectasis

has been well documented during general anesthesia and is found in as many as 90% of patients,⁴² beginning during the induction period.¹⁴⁰ Thus, unlike ALI or acute respiratory distress syndrome, there is a strong physiologic rationale supporting the use of recruitment maneuvers for reversing lung collapse during general anesthesia.

Recruitment maneuvers in anesthesia are most commonly performed by “bag squeezing,” that is, as sustained inflation, using the airway-pressure limiting valve of the anesthesia machine (see Fig. 24-2). Although simple, this maneuver has some drawbacks. First, the airway pressure is not kept constant, changing with the pressure applied on the bag by the anesthesiologist and the fresh gas flow from the anesthesia machine. Second, switching from manual to controlled ventilation usually leads to a loss of pressure in the anesthesia circuit, and the PEEP valve may require a couple of cycles to achieve the desired level. Thus, depending on the time constant of lungs and breathing circuit, derecruitment can occur with this kind of maneuver. Third, if the anesthesia ventilator initiates an inspiration immediately after switching and the airway pressure is still high, pressure overshoot with barotrauma may result.

Such limitations can be overcome by applying sustained inflations by means of continuous positive airway pressure (CPAP). Unfortunately, CPAP is not available on all anesthesia ventilators (see Table 24-1). Alternatively, the limitations of manual recruitment maneuvers can be avoided if airway pressure is increased during tidal ventilation (so-called cycling maneuver¹⁴¹), for example, with a stepwise increase of PEEP during volume-controlled ventilation (VCV), or even PCV, if this latter mode is available on the anesthesia ventilator.

Figure 24-5 illustrates two possible ways of recruiting the lungs with cycling maneuvers in the VCV mode. Initially, the intravascular volume is expanded to achieve normovolemia, and the fractional inspired oxygen concentration (FI_{O_2}) is increased to 1. Tidal volume is set at 8 to 12 mL/kg and the inspiratory pause at 20% to 50% of inspiration. In the first variant, PEEP is increased in steps of 5 cm H₂O up to 20 cm H₂O every three to five breaths. Then, the PEEP is reduced in steps of 2 to 3 cm H₂O and static elastance of the respiratory system measured at each step (decremental PEEP trial). In the second variant, the PEEP is increased to 10 to 12 cm H₂O and maintained constant thereafter. V_T is increased in steps of 2 to 4 mL/kg until an inspiratory plateau pressure of 30 to 40 cm H₂O is achieved. Finally, V_T is again set at the initial value. This maneuver has been proposed recently for recruitment of noninjured lungs within a multicenter randomized clinical trial on the use of PEEP intraoperatively.¹⁴²



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Schematic of airway pressure during two-lung recruitment maneuvers that can be performed in the volume-controlled ventilation mode on most anesthesia machines. *Top*: in the ascending part, a stepwise change of positive end-expiratory pressure (PEEP) by 5 cm H₂O at constant tidal volume is performed until the desired inspiratory plateau pressure; in the descending part, PEEP is decreased by steps of 2 to 3 cm H₂O and the respiratory system elastance measured at each step; a second recruitment maneuver must follow before the PEEP of minimal elastance is set. *Bottom*: stepwise change of tidal volume at constant PEEP until the desired inspiratory plateau pressure, as described in reference 142.

In PCV, the driving pressure is kept constant at 15 to 20 cm H₂O and the stepwise increase and decrease of PEEP is performed as for VCV. Also with this mode, the goal is to achieve an inspiratory plateau pressure of approximately 30 to 40 cm H₂O during three to five breaths.

If the anesthesia ventilator does not enable higher levels of PEEP, V_T or driving pressures may be changed to achieve the desired opening-pressure target. Obviously, this target must be adapted according to a patient's condition, for example, increased elastance of the chest wall, which will reduce the effective transpulmonary pressure. In patients with normal lungs and chest wall elastance, airway pressure of 40 cm H₂O is sufficient to recruit the lungs,¹³¹ but morbidly obese patients may require up to 60 cm H₂O.¹³⁶

Table 24-6 depicts some of the clinical studies that have addressed the effects of recruitment maneuvers during general anesthesia, including the type of maneuver used. Oxygenation was effectively improved in all but one study, mainly when combined with higher levels of PEEP. In patients undergoing cardiac surgery, PaO_2 was still higher 24 hours after surgery. In some of the trials, moreover, the FRC was increased and respiratory system mechanics improved, while a decrease in atelectasis was documented in children undergoing magnetic resonance imaging (MRI),¹⁴³ coronary artery bypass,¹⁴⁴ and bariatric¹³⁷ surgery. Recruitment maneuvers after cardiac surgery resulted in a decrease of cardiac output.¹⁴⁵ Recruitment maneuvers, however, performed during elective cardiac surgery did not lead to hemodynamic instability in another study.¹⁴⁶

Table 24-6: Randomized Clinical Trials on the Use of Recruitment Maneuvers during General Anesthesia

		Control Group		Protective Group			Intraoperative Outcomes			Postoperative Outcomes		
Ref	Type and Number Patients/Setting	PEEP	V _T mL/kg IBW	PEEP	V _T mL/kg IBW	Type of Recruitment Maneuver	compl	oxy	oxy	atelect	PPC	Length of Stay
Tusman 1999 ¹³¹	Abdominal surgery (<i>n</i> = 30)/OR	0	7/9	5	7/9	Increase in PEEP and V _T until PEEP = 15 cm H ₂ O, V _T = 18 mL/kg or PawPEAK = 40 cm H ₂ O	=	=	=	NA	=	NA
				5+R	7/9		↑	↑	↑	NA	=	NA
Dyhr 2002 ¹⁴⁴	Coronary artery bypass surgery (<i>n</i> = 16)/ICU	0+R	6	Best PEEP+R	6	20-s inflations to Paw = 45 cm H ₂ O twice	=	↑	↑	=	NA	NA
Pang 2003 ¹³³	Laparoscopic cholecystectomy (<i>n</i> = 24)/OR	0	10	5+R	10	Manually ventilated with Paw = 40 cm H ₂ O for 10 breaths (1 min)	NA	↑	=	NA	NA	NA
Tusman 2003 ¹⁴³	Children (age 6 mo to 6 y) undergoing MRI in general anesthesia (<i>n</i> = 24)	0	Spont breath	5	Spont breath	Manually ventilating with PawPEAK = 40 cm H ₂ O and PEEP = 15 cm H ₂ O for 10 breaths	NA	NA	NA	=	NA	NA
				5+R	Spont breath					↓		
Whalen 2006 ¹³⁷	Laparoscopic bariatric surgery (<i>n</i> = 20)/OR	4	8	12+R	8	Increasing PEEP in a stepwise fashion: 10 (3 breaths), then 15 (3 breaths), finally 20 (10 breaths); PawPEAK ≥50 cm H ₂ O	↑	↑	=	NA	=	=
Celebi 2008 ¹⁴⁷	Coronary artery bypass surgery (<i>n</i> = 60)/ICU	5	7	5+R	7	R with CPAP: 30 s with Paw = 40 cm H ₂ O; R+PEEP: ↑PEEP = 20 cm H ₂ O and ↑ V _T until Paw = 40 cm H ₂ O for 2 min; after R in both groups PEEP ↓ in 1 to 2 cm H ₂ O every 5 min until the lowest PEEP (>5 cm H ₂ O) providing best	↑	↑	↑	↓(evaluated on atelect score)	↓	↓
				5+RR	7		↑	↑	↑	↓(evaluated on atelect score)	↓	↓
Chalhoub 2007 ¹⁴⁸	Bariatric surgery (<i>n</i> = 52)/OR	8	10	8+R	10	Paw = 40 cm H ₂ O for 15 s	NA	↑	↑	NA	NA	NA

Ref	Type and Number Patients/Setting	Control Group		Protective Group			Intraoperative Outcomes			Postoperative Outcomes		
		PEEP	V _T mL/kg IBW	PEEP	V _T mL/kg IBW	Type of Recruitment Maneuver	compl	oxy	oxy	atelect	PPC	Length of Stay
Minkovich 2007 ¹⁴⁶	Elective cardiac surgery (<i>n</i> = 95)/OR-ICU	5	8/10	5+R	8/10	Paw = 35 cm H ₂ O for 15 s before cardiopulmonary bypass and Paw = 30 cm H ₂ O for 5 s before ICU admission; R repeated after 20 to 30 min ICU admission	NA	↑	↑	NA	NA	=
Almarakbi 2009 ¹⁴⁹	Laparoscopic bariatric surgery (<i>n</i> = 60)/OR	10	10	10+R	10	R with Paw = 40 cm H ₂ O for 15 s; in RR+PEEP groups Paw = 40 cm H ₂ O for 15 s repeated every 10 min	↑	↑	↑	NA	=	↓
		0+R	10	10+RR	10		↑↑	↑↑	↑↑		=	↓↓
Reinius 2009 ¹³⁶	Bariatric surgery (<i>n</i> = 30)/OR	0+R	10	10	10	Paw increased to 55 cm H ₂ O, and an inspiratory hold was kept for 10 s	↑	↑	=	=	NA	NA

Abbreviations: ↑, increased; ↓, decreased; =, unchanged; *atelect*, atelectasis; *compl*, compliance; *IBW*, ideal body weight; *ICU*, intensive care unit; *MRI*, magnetic resonance imaging; *NA*, not assessed; *OR*, operation room; *oxy*, arterial oxygenation; *PaO₂*, partial pressure of arterial oxygen; *Paw*, airway pressure; *Paw_{PEAK}*, peak airway pressure; *PEEP*, positive end-expiratory pressure; *PPC*, postoperative pulmonary complication; *R*, recruitment maneuver; *RR*, respiratory rate; *V_T*, tidal volume.

As can be inferred from those trials, recruitment maneuvers are effective in reversing atelectasis, increasing FRC and improving oxygenation in general anesthesia, with little or no effect on hemodynamics. Their impact on clinical outcome, however, is not established and, therefore, their routine use cannot be recommended for unselected patient populations.

Spontaneous and Assisted Spontaneous Breathing

Most anesthetic agents are depressants of the respiratory center and some have additional muscle relaxant properties, most notably the volatile anesthetics. Because induction of general anesthesia is conducted with comparatively high dosages of anesthetic drugs, apnea usually develops. Furthermore, endotracheal intubation is frequently needed during general anesthesia to protect the airways from aspiration of gastric contents. To facilitate laryngoscopy and placement of an endotracheal tube, muscle paralytic agents are used. In view of these facts, spontaneous breathing activity is frequently abolished following induction of general anesthesia, and controlled mechanical ventilation is required in order to maintain an adequate gas exchange.

Nevertheless, endotracheal intubation itself can result in different complications, including injuries of the teeth, vocal cords, and arytenoids, which may be avoided if the patient is not at risk of aspiration. For example, supraglottic devices, and more specifically laryngeal masks, can maintain the patency of the upper airways and facilitate ventilation by spontaneous breathing or positive pressure ventilation. Furthermore, short anesthesia procedures may be conducted with a facemask, with or without placement of a Guedel tube, in a spontaneously breathing patient. Even if the use of an endotracheal tube is necessary, general anesthesia can be titrated to allow spontaneous breathing activity, mainly when combined with regional anesthesia techniques. Thus, maintenance of adequate gas exchange during general anesthesia may be possible by spontaneous breathing alone or, alternatively, with assisted ventilation, if tidal volume has to be increased or the work of breathing reduced.

Possible Advantages

General anesthesia leads to a decrease in FRC and development of atelectasis in dependent lung zones, which may, in turn, impair gas exchange. Spontaneous breathing activity can counteract such effects if enough transpulmonary pressure is generated in dependent lung zones.¹⁵⁰ Even in the absence of alveolar lung recruitment, spontaneous breathing activity may improve gas exchange by facilitating the redistribution of perfusion towards better aerated and ventilated ventral areas.¹⁵¹ Also, inspiratory efforts may reduce the mean intrathoracic pressure, contributing to an increase in venous return to the heart and the perfusion of different organs.¹⁵² Furthermore, the maintenance of spontaneous breathing during anesthesia may also have beneficial effects on the lung tissue. In healthy rats

under general anesthesia, controlled mechanical ventilation elicits a degradation of glycosaminoglycans of the extracellular matrix and promotes edema formation, which can be prevented by spontaneous breathing.⁸⁸

The presence of spontaneous breathing during general anesthesia is usually associated with a reduction in the use of anesthetic drugs and obviates the need for neuromuscular blocking agents, which may speed the recovery from anesthesia and contribute to avoid complications from inadvertent residual sedation and/or muscle paralysis.¹⁵³ Furthermore, a reduction in the use of anesthetic agents will increase the levels of stress hormones, for example, adrenaline, noradrenaline, adrenocorticotrophic hormone, and cortisol, decreasing the requirements for exogenous vasoactive drugs.

Possible Disadvantages

Some anesthesiologists fear that the decrease in depth of anesthesia for maintaining spontaneous breathing activity may result in intraoperative awareness and recall. Indeed, anesthesia depth requirements may vary according to the surgical stimulus and maintenance of adequate analgesia while avoiding respiratory depression can be challenging. Also, spontaneous breathing activity may be insufficient to generate adequate alveolar ventilation or result in increased work of breathing secondary to a relatively low diameter of the airway assist device, requiring assistance by positive pressure. When pressure-support ventilation, which is the most common mode of assisted mechanical ventilation in anesthesia ventilators, is used for assisting spontaneous breathing, not enough transpulmonary pressure is generated in dependent zones to recruit the lungs.¹⁵⁴ Furthermore, other modes of assisted ventilation also available on anesthesia machines, for example, synchronized intermittent mandatory ventilation, may result in patient-ventilator asynchrony, leading to patient discomfort and even arousal.

Nonconventional Modes of Mechanical Ventilation during Anesthesia

In contrast to the ICU, where nonconventional modes of mechanical ventilation are used for severe impairment of gas exchange, respiratory system mechanics, or both, patients under general anesthesia may require differentiated ventilator strategies as a means to facilitate the surgical approach or diagnostic procedures. High-frequency ventilation modes, especially high-frequency jet ventilation (HFJV) and high-frequency oscillatory ventilation (HFOV), are particularly interesting for the anesthesiologist, because they achieve adequate gas exchange while limiting lung and chest excursion, or saving space in the larynx for placement of surgical and diagnostic tools. HFJV is also helpful in emergency situations where supraglottic obstruction is present and percutaneous transtracheal ventilation through a small-bore catheter becomes necessary.

The devices used for high-frequency ventilation during general anesthesia are virtually the same as those found in the ICU. Obviously, those systems do not support use of volatile anesthetics and anesthesia is performed with intravenous drugs.

High-Frequency Jet Ventilation

The possible indications for HFJV include: (a) surgery and diagnostic procedures of the larynx and trachea (e.g., endoscopy, placement of tracheal stents, laser surgery); (b) improvement of oxygenation during one-lung anesthesia; and (c) transtracheal oxygenation in emergency situations. This ventilation mode, however, is contraindicated in patients who are at risk for aspiration and with upper airway obstruction.

HFJV can be delivered either through a tube or cannula placed supraglottically or subglottically. With the supraglottic approach, the jet ventilator can be attached to an endoscope; with the subglottic approach, either a transtracheal or a transglottic route can be used. In the transtracheal variant, a 13-gauge to 14-gauge (adults) or 18-gauge (children younger than age 3 years) catheter is introduced through the cricothyroid membrane into the trachea using the Seldinger method or under endoscopic control. The catheter tip is then advanced and placed 2 to 4 cm above the carina.¹⁵⁵ In the transglottic variant, a catheter with an internal diameter of 1.5 to 3.0 mm and up to 30 cm length is introduced transnasally or transorally and its multipore tip is advanced into the trachea under direct visualization by laryngoscopy using intubation-assist devices.

Jet ventilators apply pressures in the range of 1.5 to 2.5 bars with a respiratory frequency of 100 to 150 breaths/min and inspiration time of 50% (30% to 70%) on the jet catheter.⁷⁵ These settings result in airway pressures lower than those obtained with conventional mechanical ventilation, provided there is no stenosis or obstruction of the upper airways. Also, those settings result in tidal volumes of 1 to 3 mL/kg, which can be increased by augmenting the pressure, reducing the respiratory frequency, and increasing the duration of inspiration. Accordingly, if the tidal volume has to be reduced, for example, during arousal from anesthesia, so as to stimulate spontaneous breathing, the respiratory frequency is increased up to 300 breaths/min. Inspiratory oxygen fractions during HFJV are in the range of 50% to 100%, but the effective oxygen fraction in the trachea is usually lower because of air entrainment,¹⁵⁶ that is, mixing of oxygen from the ventilator with air surrounding the jet catheter by means of the Venturi effect.

Although adequacy of oxygenation can be monitored by usual means, for example, noninvasive pulse oximetry, capnography is not suitable for accessing the efficiency of ventilation during HFJV. Thus, transcutaneous or partial pressure of arterial carbon dioxide (P_{CO_2}) measurement may be required for appropriate titration of HFJV. Furthermore, the measurement of the effectively delivered tidal volume during HFJV is not possible in clinical practice, although it can be performed by body plethysmography in the lab. In contrast, airway pressure can be monitored by some jet ventilators through an accessory route in double-lumen jet catheters, or during end-expiratory pauses in conventional single-lumen jet catheters.

Serious complications from HFJV are rare.¹⁵⁵ Pneumothorax and pneumomediastinum have been reported in less than 1% of patients,^{155,157} while subcutaneous emphysema has been observed in approximately 8% of patients.¹⁵⁵ Such complications result from pulmonary hyperinflation in presence of obstruction to expiratory flow and may occur even with devices that control tracheal pressure. Newer jet ventilators offer active expiratory assistance that could minimize such

events. Supraglottic devices can theoretically promote transport of particles and liquid into the trachea, including tumor tissue, and also cause gastric inflation and rupture.¹⁵⁸ Also, when gas from the jet ventilator is not humidified, drying of the trachea mucosa and tracheobronchitis may occur.¹⁵⁹

High-Frequency Oscillatory Ventilation

Chapter 19 explains the principles of HFOV in detail. In brief, during HFOV a diaphragm or piston pump generates quasisinusoidal oscillations in airway pressure. This technique has been described originally with oscillation frequencies as high as 50 Hz,¹⁶⁰ but nowadays frequencies of 5 to 15 Hz are used clinically. To remove CO₂ from the respiratory circuit, an inspiratory bias flow of humidified and warmed air is used. The inspiratory bias flow and a gas outflow valve, which regulates the resistance of the expiratory limb, determine the mean airway pressure (mPaw).¹⁶¹ The amplitude of pressure oscillation (ΔP), which is usually in the range of 60 to 90 cm H₂O, determines the effectively applied V_T , and an inspiratory-to-expiratory timing (I:E) ratio of 1:2 to 1:1 is set separately. Although the ΔP in the HFOV is high, the downstream pressure in the airways is obviously much lower,¹⁶² and the resulting V_T is only a fraction of that obtained with conventional mechanical ventilation, in the range of 1 to 3 mL/kg. The major factors that determine the V_T during HFOV are the diameter and length of the endotracheal tube, ΔP itself, inspiratory time, impedance of the respiratory system and the oscillation frequency, whereby higher frequencies result in lower V_T .¹⁶³

The possibility of combining the lowest possible V_T with relatively high mPaw, and thereby reduce volutrauma and atelectrauma, respectively, makes HFOV particularly attractive for lung-protective ventilation. HFOV has become popular for the ventilator management of acute pulmonary dysfunction in infants, even if there is no clear evidence that it is superior to conventional ventilation.^{164,165} In the operating room, HFOV may be a valuable alternative to limit the tidal excursion of the lungs, while allowing expansion and counteracting increases in abdominal pressure that would otherwise lead to shrinkage of the lungs. Accordingly, possible indications of HFOV in pediatric anesthesia include surgery for⁷⁶ congenital diaphragmatic hernia, congenital cystic adenomatoid malformation, esophageal atresia, patent ductus arteriosus, abdominal wall defect, and enterocolitis.

The use of HFOV during general anesthesia of adult patients is practically restricted to cases where HFOV has been initiated in the ICU and continued in the operating room. Possible reasons for that restriction are the time required for stabilization of lung recruitment and gas exchange after beginning of HFOV, lack of experience of anesthesiologists with this mode, and the fact that HFOV is currently not available on anesthesia machines, precluding its use with volatile anesthetics.

Special Situations

Mechanical Ventilation during Laparoscopy

Laparoscopy is widely used in the surgical treatment of a number of diseases. Its advantages are generally believed to lie with its minimal invasiveness, better cosmetic outcome, and shorter length of hospital stay based on surgical expertise and state-of-the-art equipment. The insufflation of the abdomen, however, might cause derangements of respiratory mechanics, lung volumes, and hemodynamics. Furthermore, laparoscopy may be associated with longer operation times.

Most patients tolerate mechanical ventilation during laparoscopy without major problems. Pneumoperitoneum, however, with increased intraabdominal pressure may pose difficulties for mechanical ventilation in patients with underlying lung disease and higher-than-normal body mass index.¹⁶⁶

Abdominal insufflation of gas during laparoscopy, usually within the range of 10 to 15 mm Hg, greatly affects the respiratory mechanics, promoting an upward shift of the diaphragm and reducing lung volume.

Lung Volumes at End Expiration.

The reduction in lung volume depends not only on intraabdominal pressure but also on the position of the patient. It is expected the Trendelenburg position will result in further reduction in lung volume, reducing FRC.

Lung and Chest Wall Mechanics.

Insufflation of gas into the abdomen causes a rightward shift of the pressure-volume curve of the respiratory system. Such changes reflect the increase in both the lung and chest wall components of elastance. Also, the decrease in lung volume narrows the airways, increasing lung resistance. These effects have profound impact on the correct interpretation of the pressures displayed on the ventilator.

Contemporary anesthesia ventilators assess peak and plateau airway pressures, thus allowing discrimination between dissipation of pressure secondary to airway resistance (difference between peak and inspiratory plateau pressure) and the elastic properties of the respiratory system (inspiratory plateau pressure minus PEEP).

In the presence of abnormal chest wall mechanics, however, airway pressure does not adequately represent the transpulmonary pressure. Thus, it is necessary to closely monitor airway pressure before initiation of pneumoperitoneum, so as to set a new reference point of airway pressures that takes into consideration changes of lung and chest wall elastance, airway narrowing, and lung distortion.

Setting the Mechanical Ventilator for Laparoscopy.

During laparoscopic surgery, CO₂ is absorbed from pneumoperitoneum, increasing CO₂ content in blood. Thus, to keep a normal PaCO₂, an increase in minute ventilation of approximately 30% is needed, which can be accomplished by increasing tidal volume, respiratory rate, or both. An increase in respiratory rate,

however, in patients with expiratory flow limitation can worsen the auto-PEEP, whose monitoring may be difficult with ventilators that do not allow an end-expiratory hold or do not display flow-versus-time curves.

Arterial oxygenation is barely impaired during general anesthesia for laparoscopic surgery, even in morbidly obese patients.¹⁶⁷ Nonetheless, tissue oxygenation during laparoscopic surgery is lower than during open surgery.¹⁶⁸ The use of PEEP during laparoscopy increases PaO_2 but such effect is only marginal, independently of body weight.¹³⁴ In obese patients in beach-chair position, however, the improvement in arterial oxygenation induced by PEEP may be significant,¹⁶⁹ but higher V_T does not seem to be useful for increasing PaO_2 in such patients.¹⁷⁰

Table 24-7 shows a list of randomized controlled trials on PEEP during laparoscopy. In most studies, the effects of zero end-expiratory pressure versus PEEP of 6 to 10 cm H_2O , with^{133,137,149} or without^{134,135,138} recruitment maneuvers, and V_T of 8 to 10 mL/kg were compared. All studies showed an increase in PaO_2 intraoperatively, while two investigations reported an increase in respiratory compliance.^{137,149} In one study,¹³⁷ there were no significant differences in PaO_2 postoperatively, length of stay, and postoperative pulmonary complications when comparing low versus high PEEP combined recruitment maneuvers. Some investigators, however, reported a reduction in postoperative atelectasis and postoperative pulmonary complications with high, compared to low, PEEP.¹³⁸ With regard to hemodynamic side effects, use of PEEP during pneumoperitoneum has produced conflicting results. Although some authors found that cardiac output was decreased during pneumoperitoneum combined with high PEEP,^{137,171} others did not observe such deterioration.^{134,172}

Table 24-7: Randomized Clinical Trial Comparisons of Different Ventilator Strategies during Laparoscopy

Ref	Type and Number Patients/Setting	Control Group		Protective Group		Type of Recruitment Maneuver	Intraoperative Outcomes		Postoperative Outcomes			
		PEEP	V _T mL/kg IBW	PEEP	V _T mL/kg IBW		compl	oxy	oxy	atelect	PPC	Length of Stay
Pang 2003 ¹³³	Laparoscopic cholecystectomy (n = 24)/OR	0	10	5+R	10	Manually ventilated with Paw = 40 cm H ₂ O for 10 breaths (1 min)	NA	↑	=	NA	NA	NA
Meininger 2005 ¹³⁴	Laparoscopic surgery (n = 20)/OR	0	—	5	—		NA	↑	=	NA	NA	NA
Whalen 2006 ¹³⁷	Laparoscopic bariatric surgery (n = 20)/OR	4	8	12+R	8	Increasing PEEP in a stepwise fashion: 10 (3 breaths), then 15 (3 breaths), finally 20 (10 breaths); PawPEAK ≥ 50 cm H ₂ O	↑	↑	=	NA	=	=
Talab 2009 ¹³⁸	Laparoscopic bariatric surgery (n = 66)/OR	0	8 to 10	5	8 to 10		NA	=	=	↓	↓	=
				10	8 to 10			↑	↑	↓↓	↓↓	↓
Almarakbi 2009 ¹⁴⁹	Laparoscopic bariatric surgery (n = 60)/OR	10	10	10+R	10	R with Paw = 40 cm H ₂ O for 15 s; in R+PEEP groups Paw = 40 cm H ₂ O for 15 s repeated every 10 min	↑	↑	↑	NA	=	↓
		0+R	10	10+R	10		↑↑	↑↑	↑↑		=	↓↓
Park 2009 ¹⁷³	Laparoscopic hysterectomy (n = 50)/OR	0	10	15+R	10	10 manual breaths with PawPEAK = 40 cm H ₂ O	=	↑	↑	NA	NA	NA
Kim 2010 ¹³⁵	Laparoscopic cholecystectomy (n = 30)/OR	0	8	5	8		=	↑	↑	NA	NA	NA

Abbreviations: ↑, increased; ↓, decreased; =, unchanged; *atelect*, atelectasis; *compl*, compliance; *IBW*, ideal body weight; *ICU*, intensive care unit; *NA*, not assessed; *OR*, operation room; *oxy*, arterial oxygenation; *Paw*, airway pressure; *PawPEAK*, peak airway pressure; *PEEP*, positive end-expiratory pressure; *PPC*, postoperative pulmonary complication; *R*, recruitment maneuver; *RR*, respiratory rate; *V_T*, tidal volume.

Mechanical Ventilation Mode and Laparoscopy.

Because pneumoperitoneum can markedly affect compliance and resistance, it can also adversely affect tidal volume, minute ventilation, and alveolar ventilation during **PCV**. Thus, VCV is a valid choice to keep alveolar ventilation constant in the presence of changing lung mechanics. Most interest in the use of **PCV** during pneumoperitoneum is related to its potential for reducing peak airway pressures at similar levels of alveolar ventilation. In VCV, if end-inspiratory airway pressure does not equilibrate with alveolar pressure, transpulmonary pressure will be considerably less than the set value. Accordingly, for a given inspiratory time and peak target pressure, **PCV** applies greater cumulative pressure to the respiratory system than VCV, theoretically providing better oxygenation. Despite these theoretical advantages, however, **PCV** and VCV resulted in comparable oxygenation and respiratory mechanics during laparoscopy,^{174,175} but CO₂ elimination was more effective with VCV than **PCV** in the morbidly obese patients.¹⁷⁵

Patient Positioning and Laparoscopy.

Patient positioning is extremely important during mechanical ventilation and laparoscopy. In morbidly obese patients under laparoscopy,¹⁶⁹ reverse Trendelenburg improved end-expiratory lung volume both before and after pneumoperitoneum induction and was associated with improved arterial oxygenation. Most importantly, head-up positioning alone or the application of PEEP in a supine position produced similar effects on lung volume and oxygenation. Airway pressures, however, were much lower during the beach-chair position. These data suggest that the combination of beach-chair position with PEEP, when feasible, improves oxygenation and favors a protective lung strategy.

Mechanical Ventilation of Obese Patients

The prevalence of adult obesity has increased in the last decade in most countries,¹⁷⁶ reaching up to 35% in North America and 15% to 20% in Europe.¹⁷⁷ Because these patients show several systemic pathophysiologic alterations, intraoperative mechanical ventilation and perioperative respiratory care may be challenging to the anesthesiologist.

After induction of anesthesia in obese patients, FRC decreases to approximately 50% of preanesthesia values,^{136,178} and such change correlates well with body mass index.¹⁷⁹ The mechanism leading to reduction in FRC, with consequent development of atelectasis, is enhanced in obese patients, secondary to a possible increase in intraabdominal pressure.³⁸ Because of gravity, increased intraabdominal pressure is mainly directed toward the most dependent lung regions, leading to further reduction in diaphragmatic excursion and/or cranial displacement. Such changes in the obese patient favor the development of pulmonary atelectasis and ventilation-perfusion mismatching, which combined with anesthesia and muscle paralysis worsens arterial oxygenation.¹⁷⁹

The loss in FRC also promotes an increase in lung elastance, while increased intraabdominal pressure promotes a further increase in chest wall elastance, resulting in higher elastance of the total respiratory system. Additionally, the lung resistance is increased in obesity,¹⁷⁹ mainly secondary to the reduction in lung volume, although specific resistance remains relatively normal.⁶²

Preoperative Measures.

It is beyond the scope of this chapter to describe the innumerable particularities concerning the induction of anesthesia in the obese patient, especially those concerning difficult intubation and risk of aspiration. Some aspects, however, are relevant for mechanical ventilation following induction.

In morbidly obese patients, the time to desaturate following standard preoxygenation with pure oxygen at barometric pressure conditions is reduced. The use of CPAP or noninvasive positive pressure ventilation during induction of anesthesia can minimize the loss in FRC, preventing formation of atelectasis and increasing the duration of non-hypoxic apnea.^{180,181} In fact, use of noninvasive positive pressure ventilation before intubation improves oxygenation in obese patients undergoing different kinds of surgical approaches,^{79,182} and contributes to better preservation of FRC during laparoscopic surgery.¹⁸³ Furthermore, induction of anesthesia in obese patients in the Trendelenburg position may additionally contribute to increase the FRC.

Intraoperative Mechanical Ventilation.

In the intraoperative period, the following strategies should be considered for mechanical ventilation of the obese patient: (a) the lowest possible inspiratory oxygen fraction to achieve peripheral oxygenation saturation greater than 92%;⁵² (b) tidal volumes less than 13 mL/kg (ideal body weight);¹⁸⁴ (c) a recruitment maneuver;^{50,136} and (d) PEEP equal to or greater than 10 cm H₂O. Additionally, the use of large, manually or automatically periodically lung inflations (sighs) can be useful.⁵²

Inspired oxygen fractions greater than 0.8 should be avoided in obese patients, because they favor the formation of progressive reabsorption atelectasis in the presence of zones with a low ventilation-perfusion ratio. The use high tidal volumes (>13 mL/kg ideal body weight) does not result in further improvements in oxygenation, although it can induce hypocapnia if the respiratory rate does not decrease. Moreover, the continuous use of high tidal volumes even during anaesthesia can promote ventilator-induced lung injury and negative effects on hemodynamics.

Adequate opening pressures can be applied by periodic lung inflations (recruitment maneuvers).¹³⁶ Airway pressures up to 60 cm H₂O, however, may be necessary to achieve a transpulmonary pressure that reopens collapsed lung areas. In fact, obese patients have an increased abdominal load secondary to increased intraabdominal pressure as well as thoracic fat mass, with resulting in an increase in chest wall elastance. In these circumstances, transpulmonary pressure is reduced, making the recruitment maneuver less effective. The duration of a recruitment maneuver should be limited to a maximum of 10 seconds so as to minimize as much as possible the negative effects on hemodynamics. Importantly, recruitment maneuvers should be performed only in patients who have stable volume and hemodynamic conditions. In the absence of PEEP, recruitment maneuvers can be repeated every 30 minutes to reopen atelectasis. PEEP levels 10 cm H₂O after a recruitment maneuver, however, may prevent small airways from collapsing. Obviously, excessive PEEP may lead to negative effects, like compression of lung capillaries with increased ventilation-perfusion ratios, alveolar overdistension and reduction of cardiac output. Recruitment maneuver and higher PEEP have also proven effective to improving respiratory function during laparoscopy.⁷⁹

Postoperative Measures.

The respiratory function of obese patients is considerably impaired in the postoperative period.¹⁸⁵ Such impairment is mainly related to the postoperative pulmonary restrictive syndrome, which can lead to secondary hypoxemia.^{186,187}

A number of measures to reduce postoperative pulmonary complications have been proposed, including chest physiotherapy, patient positioning (30 to 45 degrees semirecumbent position), incentive spirometry, and noninvasive respiratory support.^{187–191} The main goal is to restore lung volumes to the preoperative values, improving oxygenation and reducing work of breathing.¹⁹² In this context, CPAP may be useful, especially when it is used before important deterioration of oxygenation.

Mechanical Ventilation of Patients with Chronic Obstructive Pulmonary Disease and Asthma

Bronchial asthma and COPD are the most common obstructive disorders of the lung. The prevalence of asthma is increasing worldwide, but improvements in medical care have led to a progressive decrease in morbidity and mortality.¹⁹³ Conversely, the prevalence of COPD is widely variable among the different regions of the world, and although different medical treatments of COPD have been developed, mortality is high and expected to occupy the second or third place among all causes of deaths within the next decade.¹⁹⁴

The common feature of bronchial asthma and COPD is an increase of airway resistance, which represents a challenge for anesthesiologists. In addition to basic drug therapy, patients with asthma or COPD require different ventilation strategies for perioperative management.^{195,196}

Various pathophysiologic mechanisms may contribute differently to airway narrowing in bronchial asthma and COPD. In asthma, acute episodes of bronchoconstriction are mainly secondary to an exaggerated response of airway smooth muscle to constrictor stimuli; in COPD, persistent airway narrowing is believed to be the result of airway wall thickening secondary to chronic bronchitis, and loss of lung elastic recoil secondary to emphysema.¹⁹⁷

Airway narrowing has three major consequences, especially during acute exacerbations of bronchial asthma or COPD, that are relevant for the management of mechanical ventilation: dynamic hyperinflation, inefficient gas exchange, and cardiovascular abnormalities.

Mechanical ventilation may affect each of these features, though significant interactions exist between them.¹⁹⁸

Dynamic Hyperinflation.

Lung hyperinflation indicates an increase of any absolute lung volume, for example, FRC, above its predicted value. This may have two main causes: first, an increase of the relaxation volume of the respiratory system (static hyperinflation); second, failure to reach relaxation volume at end-expiration (dynamic hyperinflation). The former is caused by loss of elastic recoil (emphysema) and is not amenable to current pharmacologic treatments. The latter is the consequence of increased expiratory resistance and is mechanically characterized by an elastic threshold load, so-called intrinsic positive end-expiratory pressure (PEEPi), which must be overcome before inspiratory flow can be generated.

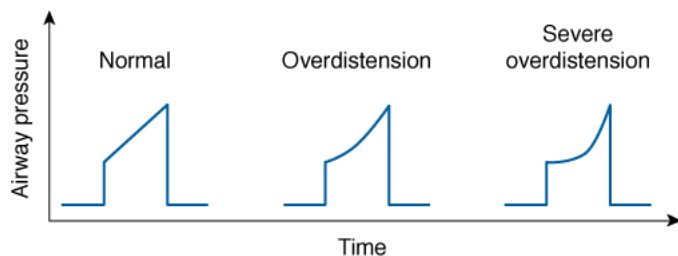
During mechanical ventilation, the occurrence of PEEPi can be caused by several mechanisms including (a) increased time-constant (product of resistance and compliance), (b) increase in breathing frequency or decrease in expiratory time, (c) an increase in V_T , and (d) airflow limitation during tidal expiration. In cases *a* to *c*, the airways remain open during the whole breathing cycle, although their diameter may be decreased; in case *d*, the airways collapse at end-expiration. The latter can be easily inferred from flow-time, pressure-time, and flow-volumes curves. On flow-time curves, flow at end-expiration does not equal zero, even when the pressure-time curve indicates that airway pressure has reached the level of set external PEEP. On the flow-volume curve, the expiratory limb of the loop shows a curvilinear shape, and it appears truncated when expiratory flow persists at end-expiration. It is worth noting that: (a) in the case of airway narrowing without achieving flow limitation, one should expect to see the following: (i) flow at end-expiration on the flow-time curve, (ii) increased pressure at end-expiratory occlusion on the pressure-time curve, (iii) a linear flow-volume curve, with a truncated flow-volume loop; and (b) in the case of airflow-limitation, the only difference is the presence of a curvilinear (convex toward the volume axis) flow-volume curve.

When PEEPi occurs secondary to mechanisms other than expiratory flow limitation, application of external PEEP produces a further increase in PEEPi and a plateau pressure of the respiratory system, with possible negative hemodynamic effects. Conversely, application of external PEEP in the presence of flow limitation will not affect plateau pressure and hemodynamics, provided it is limited to equal to or less than 75% of PEEPi.

For practical purposes, continuous displays of ventilator waveforms can assist the clinician in detecting and monitoring pathophysiologic changes, optimize ventilator settings and treatment, determine effectiveness of ventilator settings, and minimize the risk of ventilator-induced complications. Flow and volume waveforms can be employed to directly estimate end-expiratory volume above passive FRC. The volume expired during a prolonged apnea (up to 40 seconds) may be used to determine the volume of gas above FRC. The total exhaled volume is measured from the end of inspiration until there is no visually detectable change in volume.

Monitoring of pressure-time curves can also be helpful in avoiding excessive hyperinflation (Fig. 24-6). In the case of normal inflation, the pressure-time curve during inflation at constant flow in VCV is linear (i.e., pressure increases in proportion to an increase in volume); in the case of overdistension, a convexity toward the time axis appears (i.e., a progressive increase in pressure that is not proportional to increase in volume).

Figure 24-6



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Schematic of airway pressure versus time curves in a normal subject and in patients with lung overdistension and severe overdistension. The degree of curvature can be related to the severity of overdistension, but is not always present.

In summary, the monitoring of pressure, flow, and volume over time associated with end-inspiratory and end-expiratory occlusion maneuvers is essential to better define the presence and causes of dynamic hyperinflation.

Gas Exchange.

Inefficient gas exchange is signaled by $\dot{V}_A/\dot{Q}$ hypercapnia or hypoxemia. Hypoxemia of variable degree, caused mainly by mismatching (mainly venous admixture), is almost invariably present in COPD and during asthma attacks. Hypercapnia reflects both $\dot{V}_A/\dot{Q}$ mismatching and alveolar hypoventilation, the latter resulting from both respiratory muscle dysfunction and increased ventilatory requirements. During asthma attacks, hypoxia is generally associated with hypocapnia, and increasing values of Pa_{CO_2} are a warning sign of potential fatality secondary to respiratory arrest.

Cardiovascular Abnormalities.

Cardiovascular dysfunction is usually related to acute and chronic blood-gas derangement, dynamic hyperinflation, and increased right-ventricular afterload. Increases in pulmonary artery pressures may result from hypoxic pulmonary vasoconstriction mainly in hypoxemic patients with COPD, and also from low-grade systemic inflammation as evidenced by elevated levels of serum C-reactive protein and tumor necrosis factor. Left-heart failure may also develop in hyperinflated patients with COPD because of an altered mechanical interaction between heart and lung. Nevertheless, the association between COPD and left-ventricular dysfunction may be simply reflect old age and exposure to common risk factors, such as smoking habit.

Postoperative Pulmonary Complications.

Patients with COPD are at increased risk for postoperative atelectasis, pneumonia, and even death.¹⁹⁹ In patients with severe COPD undergoing noncardiothoracic surgery, the incidence of postoperative pulmonary complications is approximately 37% (excluding atelectasis) and the 2-year mortality is approximately 47%. A recent study, however, in cardiothoracic surgery reported that COPD is not a major independent risk factor for postoperative pulmonary complications.²⁰⁰

Risk Stratification.

In a general surgical population, independent risk factors for postoperative pulmonary complications in adults with respiratory disease include advanced age, preexisting pulmonary disease, cigarette smoking, congestive heart failure, functional dependence (e.g., the inability to perform activities of daily living), and site of surgery (with highest risks for thoracic and abdominal surgery).¹⁰² Epstein et al²⁰¹ developed a cardiopulmonary risk index that is a combination of the modified Goldman cardiac index and pulmonary risk factors [obesity, productive cough, wheezing, tobacco use, forced expiratory volume in 1 second (FEV_1)-to-forced vital capacity (FVC) ratio of $\text{Pa}_{\text{CO}_2} > 45$ mm Hg]. Those with a cardiopulmonary risk index of equal to or greater than 4 were twenty-two times more likely to develop a complication following major thoracic surgery.

In patients with severe COPD, Wong et al.¹⁹⁹ estimated the incidence of five different postoperative complications: death, pneumonia, prolonged intubation, refractory bronchospasm, and prolonged ICU stay. From multivariate analysis, they concluded that American Society for Anesthetists (ASA) physical status equal to or greater than IV, Shapiro score equal to or greater than 5, and FEV_1 were preoperative risk factors, and emergency operation, abdominal incision, anesthesia duration longer than 2 hours, and general anesthesia were intraoperative risk factors. Risk stratification of COPD patients by ASA physical status was associated with higher incidence of postoperative pneumonia, prolonged postoperative intubation, and higher mortality. Additional risk factors include the severity of COPD, assessed by spirometry and 6-minute walk distance results, and body mass index.

More recently the use of dedicated scores has been proposed. A surgical lung injury prediction model to predict risk of postoperative ALI based on readily available preoperative risk factors has been developed. It includes high-risk cardiac, vascular, or thoracic surgery, diabetes mellitus, COPD, gastroesophageal reflux disease, and alcohol abuse.¹¹⁷

Other investigators⁹⁹ did not identify COPD or bronchial asthma as individual risk factors for postoperative pulmonary complications after general anesthesia for thoracic or nonthoracic surgery. Seven independent risk factors were identified: low preoperative arterial oxygen saturation (SaO_2), acute respiratory infection during the previous month, age, preoperative anemia, upper abdominal or intrathoracic surgery, surgical duration of at least 2 hours, and emergency surgery. The factor capturing a low preoperative SaO_2 likely included patients with severe COPD.

We believe that the use of scores for better predicting postoperative pulmonary complications should be further developed to better identify patients at risk for complication in the intraoperative and postoperative period.²⁰²

Preoperative Preparation.

Preoperative management involves assessment of general physical status (pulmonary, cardiac, neurologic diseases) and treatment of any reversible signs or symptoms. As a general principle, pulmonary function should be optimized preoperatively by standard treatments. In asthmatic patients, good preoperative control of symptoms and lung function should be achieved by bronchodilator and anti-inflammatory treatment (combination treatment with long-acting β_2 -agonists and inhaled corticosteroids), also including leukotriene-receptor antagonists, to reduce the incidence of life-threatening perioperative complications possibly linked to airway hyperresponsiveness.¹⁹⁸ Additional preoperative management includes adequate control of secretions and infection and sound control of anesthesia.

Spirometry is generally used as a guide for both diagnosis and treatments of asthma and COPD, although a chest radiograph may be occasionally necessary for diagnosis.²⁰³ Arterial blood-gas measurements are performed as needed. Indicators for arterial blood-gas analysis include spirometric values of FEV₁ and FVC less than 50% of predicted, FEV₁ less than 1 L, or FVC less than 1.5 L. PaCO₂ greater than 45 mm Hg is a strong risk factor for postoperative pulmonary complications in patients with COPD; PaCO₂ values greater than 50 mm Hg are likely to require postoperative mechanical ventilation following major surgery, whereas preoperative values equal to or less than 45 mm Hg can be usually managed by controlled oxygen therapy and careful monitoring of arterial blood gases.^{204,205} Postponing elective surgery should be considered if improvement of pulmonary function can be expected to occur over extended time periods. Preoperative education regarding postoperative deep breathing, incentive spirometry, or CPAP may improve the final outcome.

General Anesthesia in Patients with Chronic Obstructive Pulmonary Disease.

Patients with chronic lung hyperinflation may actually be less prone to develop computed tomography scan evidence of dependent atelectasis during anesthesia and paralysis. In awake patients with COPD, the $\dot{V}_A/\dot{Q}$ distribution is more heterogeneous than in healthy subjects, showing zones of relatively low $\dot{V}_A$, although with the amount of intrapulmonary shunt is negligible. In addition, computed tomography scanning shows significantly larger cross-sectional thoracic areas than in subjects with healthy lungs. During anaesthesia and paralysis, patients with COPD suffer a further worsening of $\dot{V}_A/\dot{Q}$ mismatch without increase in intrapulmonary shunt. Furthermore, FRC is only minimally reduced,³⁶ which is in contrast to findings in patients with healthy lungs.²⁰⁶

Therefore, general anesthesia with controlled ventilation should be avoided whenever possible in patients with COPD, and regional anesthesia techniques are preferred if there are no contraindications.

Targets for Controlled Mechanical Ventilation.

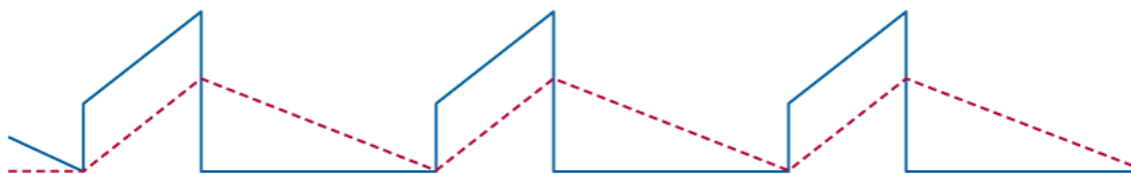
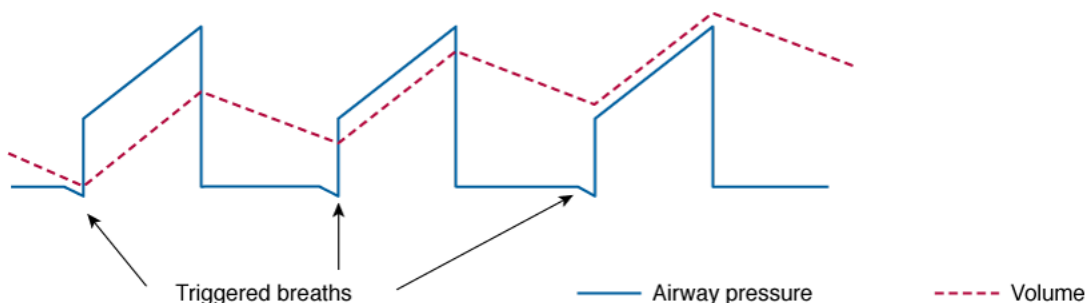
In COPD or bronchial asthma, mechanical ventilation may be difficult for the following reasons: high level of lung hyperinflation; high risk of barotrauma and volutrauma; and hemodynamic instability. Thus, minimizing the magnitude of dynamic hyperinflation during mechanical ventilation is central to the management of asthma and COPD and the following are some strategies that can be adopted:

1. Decreasing minute ventilation by reducing tidal volume, respiratory frequency, with permitted hypercapnia and mild respiratory acidosis.
2. Increasing expiratory time. In patients with asthma or COPD the time required for a complete expiration often requires 3 seconds or more, and ventilator settings that do not allow adequate time for exhalation can lead to or worsen dynamic hyperinflation.²⁰⁷ An increase in expiratory time can be achieved by increasing inspiratory flows, at the expense of increasing peak dynamic pressures, and avoiding end-inspiratory pause.
3. Reducing expiratory flow resistance through use of bronchodilators or low-density gas mixtures (e.g., 80% helium and 20% O₂), which may help reduce dynamic hyperinflation.

We recommend, whenever possible, to simultaneously apply all of these strategies. As a starting point for ventilating patients with COPD or severe asthma, we recommend that the ventilator be used in the PCV mode, setting the pressure to achieve a tidal volume of 6 to 8 mL/kg, respiratory rate of 11 to 14 breaths/min, and PEEP at 0 to 5 cm H₂O. We recommend the use of these settings with a goal of obtaining a pH generally greater than 7.2 and an inspiratory plateau pressure less than 30 cm H₂O. If plateau pressure is less than 30 cm H₂O cannot be maintained, then the patient must be evaluated for causes of decreased respiratory system compliance or increased resistance (i.e., pneumothorax, misplaced endotracheal tube, pulmonary edema). In absence of one of these findings, efforts to further limit gas trapping must be considered. Administration of sodium bicarbonate to maintain a pH of 7.2 during controlled hypoventilation has been investigated in patients with COPD and/or status asthmaticus; however, no studies have demonstrated any benefit associated with bicarbonate infusion.

Lung recruitment maneuvers should be cautiously applied in presence of asthma or COPD. Furthermore, particular care should be given to avoid excessive spontaneous breathing during time-cycled ventilation, because the reduction of expiratory time may lead to an increased hyperinflation and PEEPi, as shown schematically in Figure 24-7. For these reasons, ACV should not be routinely used in asthma or COPD patients in the awakening phase from general anaesthesia, while pressure-support ventilation and others modes of mechanical ventilation not cycled by time may be useful.

Figure 24-7

Controlled mode - Preset RR 12 bpm - Preset T_I/T_{TOT} 0.33ACV – Preset RR 12 bpm – Actual RR 20 bpm - Preset T_I/T_{TOT} 0.33 – Actual T_I/T_{TOT} 0.50

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Schematic of airway pressure and volume versus time curves during mechanical ventilation with controlled (*top*) and assist-control ventilation (ACV) in a patient with expiratory flow limitation. Because of the short expiratory time in ACV, dynamic hyperinflation is exacerbated (dashed red line). ACV, assist-control ventilation; RR, respiratory rate; T_I/T_{TOT} , inspiratory to total respiratory time.

The Important Unknowns

Approximately 234 million major surgical procedures per year are performed under general anesthesia worldwide, and 1.3 million patients develop complications that result in up to 315,000 in-hospital deaths.⁹⁷ The incidence of pulmonary complications following surgery seems to be situated between 2.7%⁹⁸ and 56%,¹⁰¹ depending on individual predisposing factors, as well as type and duration of surgery.⁹⁹ Once a patient presents with a pulmonary complication, the length of hospital stay and probability of death increases significantly. It is unknown, however, whether mechanical ventilation management during general anesthesia influences patient outcome, and current data on this issue are conflicting. Given that most patients undergoing general anesthesia have no significant lung disease, it is important to identify by appropriate scores patients who will be at higher at risk of developing pulmonary complications.²⁰² The importance of selecting appropriate settings of mechanical ventilation in the intraoperative phase is unclear, and argumentation is based on physiologic and pathophysiologic conjectures and laboratory experience, rather than outcome evidence.

The Future

As discussed above, there is an increased interest in different modes of mechanical ventilation during general anesthesia. We believe that ventilator modes and strategies that have been available only for critically ill patients will become available in the operating room, including modes of assisted spontaneous breathing and high-frequency ventilation. Moreover, strategies to ventilate the lungs during general anesthesia in different conditions, including open-abdominal surgery, one-lung anesthesia, patients with obesity, asthma, COPD or in combination, and laparoscopic surgery, among others, will be investigated and their importance in the development of pulmonary complications better defined. For this purpose, multicenter clinical trials on intraoperative mechanical ventilation will be necessary.

Summary and Conclusions

Mechanical ventilation is becoming increasingly complex not only in the intraoperative phase, but throughout the whole perioperative period. Accordingly, ventilator modes that have been used almost exclusively in the ICU can now be found on anesthesia ventilators. Such development is guided by the needs to provide adequate ventilator support in a variety of situations, including open and laparoscopic surgery, diagnostic procedures of the airways, one-lung anesthesia, presence of expiratory flow limitation (e.g., asthma, COPD), and also morbid obesity. Also, strategies of protective ventilation with low V_T and higher levels of PEEP combined with lung-recruitment maneuvers, as well as invasive and noninvasive assisted spontaneous ventilation, are becoming more popular in general anesthesia, even in the absence of ALI. Whether such protective ventilation or the use of advanced modes of ventilator support during general anesthesia will contribute to reduce the incidence of pulmonary complications has yet to be determined.

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